Department of Mathematics & Applied Mathematics

MAM1000W

Semester 1, 2022

Tutorial Questions

Tutorial Questions 1

Each week, the problem sets will be based on what you did in class the week before. This week is a bit different, as there was no 'week before' at least in MAM1000W time. So, we are going to give you practice on a few of the topics that we are doing this week, as well as an introduction to some ideas about mathematical statements. If you haven't come across these concepts yet, don't worry, you will very soon!

We're going to use this problem set to introduce some ideas of using the mathematical language, for that's what it is: a language, and a way of communicating our thoughts and ideas. The beauty of mathematics is that when we do it right it can be the most precise language in the universe. When we use it wrong though, it can be confusing and misleading.

This first week we will not have the usual tutorial (online) meeting but we will put up solutions to these problems online, towards the end of the week, for you to check your working.

The next week's homework is the first homework that you will get to speak in depth with tutors about, but for this one, see how far you can go with it. If you struggle with some bits, let us know and we can point you in the right direction.

1. Solve the inequalities
   (a) $2 \leq 4x - 5 \leq 11$
   (b) $x^2 + x + 1 > 0$
   (c) $x^2 - 7x + 12 \leq 0$
   If you have trouble with these, check out example 2.1.6 of MAM1000W Resource book 1 and pages A4-A6 (the appendix) of Stewart Calculus 9E
2. Which of the following sets is the empty set?
   a) $\{x | x \text{ is an odd integer}\}$;
   b) $\{x | x \text{ is an integer and } x + 8 = 8\}$;
   c) $\{x | x \text{ is a negative integer and } x \geq 1\}$.
   If you have trouble with these, check out the section on Sets in the MAM1000W Resource book 2
3. A circular cone with height H and radius R is placed so that the point of the cone faces upwards. The top part is cut off, to form a *truncated cone*. The height of the part that is cut off is h.
   a) Draw the diagram of the resulting
   b) Show that the volume of the *truncated cone* is given by $V = \dfrac{\pi R^2}{3H^2}(H^3 - h^3)$.

   > Recall:
   > The volume of a circular cone with height h and radius r is given by $V = \frac{1}{3}\pi r^2 h$.

4. Which of the following statements are true, and which are false?

   > a) $\sqrt{9} = 3$ and -3.
   > b) $\sqrt{9} = 3$.
   > c) $\sqrt{9} \neq -3$.
   > d) If $x^2 = 9$ then $x = 3$ or $x = -3$.
   > e) $\sqrt{x^2} = x$ for all x.
   > f) $\sqrt{x^2} = x$ for some x.

   Once you have decided, *check your answer with a tutor*!

5. Are there integers a and b such that $18a + 3b = 1$
   hint: Suppose there are such integers, then what can you conclude?
6. Here are some slightly trickier inequalities to solve:
   a) $2x^2 + 4x > x^2 - x - 6$
   b) $x^3 - x^2 > 6x$
   c) $-x^2 + 6x - 9 > 0$
7. Below are three statements, and below each statement is an "argument" or "proof" which appears to show that the statement is true. In each case:

   a) say whether the statement is true or false;
   b) say whether you think the argument is correct.

   Here you have to be careful: a statement may be true, but the argument given for it being true may be wrong.

   Statement 1: If you square any odd number and divide the answer by 4, the remainder is 1.

   Argument: The number 3 is odd, and $3^2 = 9$, and if you divide 9 by 4 you get the remainder 1.

   Statement 2: All lecture rooms can seat more than 200 students.

   Argument: My MAM1000W lecture room can seat more than 200 students, so all lecture rooms can seat more than 200 students.

   Statement 3: If $(x - 1)(y - 1) = 1$ then $x = 2$ and $y = 2$.

   Argument: If you put $x = 2$, then $x - 1 = 1$; similarly, if you put $y = 2$, then $y - 1 = 1$, so $(x - 1)(y - 1) = 1$.

   Now look at each of the three statements above and

   a) if you said the statement is true and the argument is correct, do nothing;
   b) if you said the statement is true, but the argument is not correct, then give a correct argument;
   c) if you said the statement is false, give an argument to show that it is false.

   Suppose you are given a statement that claims that something is true generally (such as Statement 1, which is claimed to be true for *any* odd number, and Statement 2 which is claimed to be true for *all* lecture rooms; the phrase *for every* also appears in such statements).

   a) If you think such a statement is false, what would you have to do to convince an opponent (somebody who does not want to believe you)?

b) If you think such a statement is true, what will you have to do to convince an opponent?

Statement 3 does not contain any of the words "any", "every" or "all" . Do you think it makes a similar general claim? Give a reason for your answer.

8. Can you see why there might be some disagreement for the following statements? Hint: Think carefully about what the word "or" could mean.
   a) $\sqrt{9} = \pm 3$.
   b) $\sqrt{9} = 3$ or -3.
   This is a subtle problem

9. A list of car registration numbers and registered owners is drawn up. Do you think this list represents a function? Motivate your answer; there may in fact be more than one correct answer.
10. Consider the set $A = \{1, 0\}$, state whether each of the following statements are true or false:
    a) $\{1\} \in A$
    b) $\emptyset \in A$
    c) $\{1\} \subset A$
    d) $0 \in A$
    e) $1 \subset A$.
11. On the Island of Imperfection live three communities called the Sinusoids, the Quadratics, and the Asymptotes. The Sinusoids tell only lies, the Quadratics tell lies or truths, and the Asymptotes tell only the truth. We are introduced to three people, one from each community , called Alice, Bishop, and Matsepe, who tell us the following:
    - Alice says: "I am a Quadratic".
    - Bishop says: "Alice is an Asymptote".
    - Matsepe says: "Bishop belongs to a less truthful community than Alice".

    To which communities do Alice, Bishop, and Matsepe belong?
    A simple answer is not sufficient: You must show why your answer is correct. Is your solution the only possible one? A hint is to pretend that one of the people is, say, an Asymptote, and see what you can deduce from that. If it leads you to a contradiction, then you've discovered something. Then see what happens if you assume that the same person is a Sinusoid. Remember that your answer must be something we can <u>read</u>, so you should spend some time thinking about how to explain your solution clearly.

Tutorial Questions 2

1. For each of the statements below, say whether it is true or false. Give reasons for your answers.
   a) For every real number a,
   (i) $\sqrt{a^2} = a$ (ii) $(\sqrt{a})^2 = a$ (iii) $\sqrt[3]{a^3} = a$ (iv) $(\sqrt[3]{a})^3 = a$.
   b) For every real number x,
   (i) $|x^2| = x^2$ (ii) $|x^3| = x^3$ (iii) $2^{|x|} = |2^x|$ (iv) $|\sin(x)| = \sin|x|$
   c) For all real numbers x and y, $|x - y| = ||x| - |y||$.
   [*Hint:* Each side of the equation represents a distance.]
2. Solve i) $|2x - 7| = 4$ ii) $|x - 7| < 2$ iii) $|5x - 3| \geq 2$;
   If you have trouble with these, check out pages A6 to A8, and exercises 43-56 (on page A8 of the appendix) of Stewart Calculus 9E
3. Find all x such that $|2x - 3| < |x - 2|$ [*Hint:* Sketch the graphs of both functions on either side of the inequality].
4. Find the intersection of: i) $[1, 5)$ and $[4, \infty)$ ii) $[1, 5)$ and $[5, 8)$ ii) $[1, 5]$ and $[2, 4]$
5. Write down the domains and ranges of the following functions using interval notation. i) $f(x) = \frac{5x-2}{x-3}$ ii) $f(x) = \sqrt{(8x - 4)(x - 3)}$ iii) $f(x) = \frac{x-2}{x^2-4}$ (note that this is subtle)
6. Solve i) $|x^2 - 2x - 3| < 1$ ii) $|x - 1| < |x - 2| + |x - 3|$
7. Prove that there is no real number x such that $|x - 4| + |x - 6| \leq 1$.
8. Let $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7, 8, 9\}$, $C = \{2, 4, 6, 8\}$, $D = \{4, 5\}$, $E = \{5, 6\}$, $F = \{4, 6\}$, and let X be a set which satisfies the following conditions: $X \subset A$, $X \subset B$, and $X \not\subset C$. Determine which of the sets A, B, C, D, E, F can equal to X.
9. Given the set $\mathbb{R}$ of real numbers, let $A = \{x \in \mathbb{R} |\ 1 \leq x \leq 3\}$ and $B = \{x \in \mathbb{R} |\ 2 \leq x \leq 4\}$. Find: i) $A \cup B$ ii) $(\mathbb{R} \setminus A) \cap B$ iii) $(\mathbb{R} \setminus A) \cap (\mathbb{R} \setminus B)$ iv)$((R \setminus A) \cap B) \cup ((\mathbb{R} \setminus B) \cap A)$.
   If you have trouble with these, check back at the notes on Sets in the Maths topics resource book.
10. Using the interval notation, write down the range of the function $f(x) = \frac{2x-5|x|}{x}$
11. Let f be the function with domain $(-\infty, 2) \cup (2, \infty)$ and rule
$$f(x) = \frac{3x}{x-2}$$
    a) Prove that f is one-to-one.
    b) Prove that if b is any real number except 3, then there exists a real number a such that $f(a) = b$
12. In this question we start with the graph of quite a simple function and then do all kinds of things to it. Some of the things we do will be familiar from school, but the function we use is not so familiar.
    a) Let $S(t) = 1$ if $0 \leq t < 1$ and $S(t) = 0$ for all other real values of t. Draw the graph of S. (This is the graph that we are going to use and change. Ask a tutor to check your group's graph.) What is the domain of S? What is its range?
    b) Let $T(t) = S(t) - \frac{1}{2}$. Draw the graph of T using the same set of axes used above. What have we really done to the graph of S to get the graph of T?
    c) Let c be a constant, and $U(t) = S(t) + c$. Explain how the graph of U can be obtained from the graph of S. (If you're stuck, look again at (b).) We have tried to generalise in this question what we did for a specific value of c in (b).)
    d) Let $W(t) = S(t + 2)$. Draw the graph of S and W on a new set of axes. If you're stuck, use a table of values for t. Try to see what we have done to S here to get W.

e) Let $X(t) = S(t-2)$. Draw the graph of X in on the set of axes for your last question. (Why have we asked this question? What's the difference between this question and the last? Get your graphs for W and X checked.)

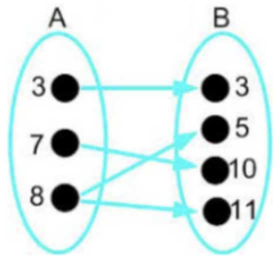
f) Let a be a constant and $Y(t) = S(t+a)$. Give a rule for obtaining the graph of Y from S. (*Hint:* (d) and (e) should help here.)

g) Let $Z(t) = S(S(t))$ (What does that mean?) Draw the graph of Z on a new set of axes. (Use tables if you are stuck.)

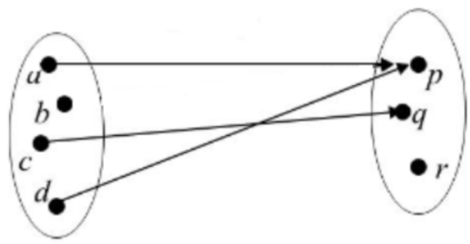
Tutorial Questions 3

1. Which of the following represents a function on a finite set? Specify the domain and range of that function.

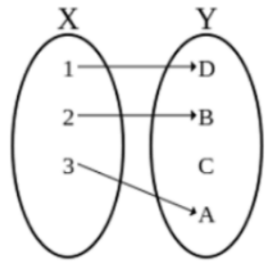
a)



b)



c)



2. Suppose that f, g are functions given on their domain by $f(x)$ and $g(x)$ respectively. For each of the following, state whether it true or false. (why?)
   a) If $f(x) = x + \sqrt{3-2x}$ and $g(u) = u + \sqrt{3-2u}$ then $f = g$.
   b) If $f(x) = \frac{x^2-3x}{x-3}$ and $g(x) = x$ then $f = g$.
3. Sketch the graph of each function given by the following equations:
   a) $f(x) = ||x| - 1|$
   b)

$$f(x) = \begin{cases} |x| & \text{if } |x| \le 1 \\ 1 & \text{if } |x| > 1. \end{cases}$$

4. Using the definition of the absolute value, solve the following inequality:

$$|x - 4| \le |2x - 3| + |x - 7|$$

5. Calculate the following by long division of polynomials:
   a) $\frac{x^4+2x-3}{x^2+x+3}$
   b) $\frac{x^3-x^2+x+1}{x+3}$
   c) $\frac{2x^4+x^2+1}{x^2-2x-1}$
   d) $\frac{x^3+x}{x-1}$
6. In this question you are going to draw a rough graph of the rational function

$$f(x) = \frac{2x^3 - x^2 - 13x + 24}{x^2 + x - 2}.$$

   a) Factorize both the numerator and the denominator (using the factor theorem if you like.)
   b) Use the factorized form of $f(x)$ to determine where it is positive and where negative. (Use a sign table.)
   c) Explain why there has to be a turning point between -2 and 1.
   d) What is the minimum number of turning points that f must have?
   e) Divide the denominator into the numerator to get f into the form of a polynomial plus a remainder term.

f) Your remainder term should be $\dfrac{-6x+18}{x^2+x-2}$. Discuss the behaviour of this remainder term for very large negative and positive x.

g) Use your answers to (f) and (g) to determine the end behaviour (the behaviour as $x \to \infty$ and as $x \to -\infty$) of f. [*Hint:* For large positive and negative x, the graph of f looks very similar to that of a much simpler function. Which one?]

h) One could argue that the graph of f has **three** asymptotes. Give them all. Does the graph of f intersect any of these asymptote? If so, where?

i) Now sketch the graph of f showing the intercepts with the axes and any slant asymptotes, and the vertical asymptotes.

7. Which of the following are odd functions, which are even, which are neither, and which are both? For each one, prove that it is so, using the definitions of odd and even functions:
   a) $f(x) = 3x^2$
   b) $f(x) = 5x^3$
   c) $f(x) = x^2 - 3x^3$
   d) $f(x) = 2^x$
   e) $f(x) = 2^{|x|}$
   f) $f(x) = 3$
   g) $f(x) = 0$

8. Suppose that a is a positive real number, that $f(x)$ is an even function, and that $g(x)$ is an odd function. Which of the following is true?
   a) $a^{f(x)}$ is an even function and $a^{g(x)}$ is an odd function.
   b) $a^{f(x)}$ is an even function but $a^{g(x)}$ is not an odd function.
   c) $a^{f(x)}$ is not an even function but $a^{g(x)}$ is an odd function.
   d) $a^{f(x)}$ is not an even function and $a^{g(x)}$ is not an odd function.

9. Let $f : \mathbb{R} \to \mathbb{R}$ be a function.
   a) Prove that the function defined by $g(x) = f(x) + f(-x)$ is even.
   b) Prove that the function defined by $h(x) = f(x) - f(-x)$ is odd.
   c) Deduce that any function $f : \mathbb{R} \to \mathbb{R}$ can be written as the sum of an even and an odd function. (Can you think of a quick way of proving this for a polynomial function?)
   d) Write the function $f(x) = e^x \sin(x + \frac{2}{3}\pi)$ as a sum of an even and an odd function

10. Consider the function $y = f(x) = \dfrac{(x+1)^2}{x^2+x-2}$.
    a) Where is the function defined?
    b) Where does the graph cut the coordinate axes?
    c) Where is the graph positive/negative?
    d) What happens near the points where the function is not defined?
    e) What happens as $x \to \pm\infty$?
    f) Does the graph cut any of its asymptotes?
    g) What is the range of the function?

    HINT: Find the values of y for which the quadratic equation in x

    $$y(x^2 + x - 2) = (x+1)^2$$

    has a solution.
    h) Sketch the graph of the function.

11. *In lectures we looked at the power functions, with non-negative integers as powers. In this question we ask you to explore the power functions where the powers are negative integers. One of the skills we would like you to acquire in this course is to draw rough graphs of functions by exploring their properties, and not relying simply on plotting points. (There is nothing wrong with plotting some points when you are trying to draw a graph of a function, but if you rely on this only, you may miss out on some important points and end up with the wrong graph.)*
Let n be a positive integer and define the function g_n by $g_n(x) = x^{-n}$. Answer the following questions. In some cases you may have to consider the cases n even and n odd separately.
    a) What is the domain of g_n?
    b) What is the range of g_n?
    c) Find $g_n(1)$ and $g_n(-1)$.
    d) What happens to $g_n(x)$ as $x \to 0$?
    e) Is g_n even, or odd, or neither?
    f) What happens to $g_n(x)$ as $x \to \infty$? And as $x \to -\infty$?
    g) Where is g_n increasing, and where is it decreasing?

    Use this information to draw, on the same system of axes, rough graphs of g_1, g_2, g_3 and g_4. Try to do this without calculating any further function values.
12. Dr Shark has observed that attendance at his first-period lecture fluctuates during the semester and suspects that it follows a predictable pattern. The highest attendance he has recorded was 103, and this happened both at his first lecture (lecture 0) and at the last lecture, and the lowest attendance was 23, at lecture number 30. There are 60 lectures in a semester. Dr Shark thinks that he can use a function of the form $f(t) = A\cos at + b$ to describe attendance, where $f(t)$ is the number of students at lecture number t, and A, a and b are constants. Below we try to do this.
    a) Begin by drawing a graph that shows attendance for one semester, putting in the information given above. Use one period of the cosine function to describe attendance for the whole semester. Put the numbers of the lectures (0 to 60) on the horizontal axis.
    b) What is the amplitude you want for your graph? Since the ordinary cosine function has amplitude 1, you will need to stretch the cosine function vertically to get the amplitude you want. How much do you need to stretch? Which constant have you just worked out?
    c) If you take an ordinary (stretched) cosine function, it oscillates about the t-axis. You want your function to oscillate around another line. Which one? To do this you will have to shift. In which direction, and by how much do you need to shift? Once you have answered these questions, you will know the value of another one of the constants. Which one, and what is its value?
    d) The period of a cosine function is 2π, and there are 60 lectures in a semester. This information enables you to find the last constant. Find it.
    e) Now write down the expression for $f(t)$, with the values of the constants filled in.
    f) Dr Shark is happy with the function you found in (e) and plans to use it in when he lectures to the same group of students in the second semester. Unfortunately, in the second semester course a new group of students with a rather different pattern of attendance joins his class. In their previous course there was a highest attendance of 75 in lecture 15 and a lowest attendance of 15 in lecture 45. Use the same method as above to find a *sine function* describing their attendance.
    g) Finally find a function that describes the attendance of the combined class. [*Hint:* The total attendance is the sum of the sum of the number of students from each group attending.] Now write this function as a single sine function of the form $h(t) = A\sin(kt + \alpha) + c$.
    h) In which lecture will the attendance in the combined class be at its lowest?

Tutorial Questions 4

1. For each of the following statements, say whether it is true or false:
   a) If f and g are linear functions, $f \circ g$ is linear.
   b) If f and g are quadratic functions, $f \circ g$ is a quadratic function.
2. Convert from degrees to radians:
   a) $210°$
   b) $9°$
   c) $-300°$
   d) $1000°$
3. Convert from radians to degrees:
   a) $\frac{5\pi}{12}$
   b) $-\frac{3\pi}{7}$
   c) 5
   d) 360
4. If $\cos(x) = -\frac{1}{3}$, where $\pi < x < \frac{3\pi}{2}$, then find:
   a) $\cot(x)$ and $\sec(x)$
   b) $\sin(2x)$
   c) $\cos(2x)$
5. a) Let $f(x) = x^2 + \frac{2}{x}$, $g(x) = \frac{2}{2x+3}$ and $h(x) = \sqrt{2x}$. Find:

$$(h \circ g \circ f)(4)$$

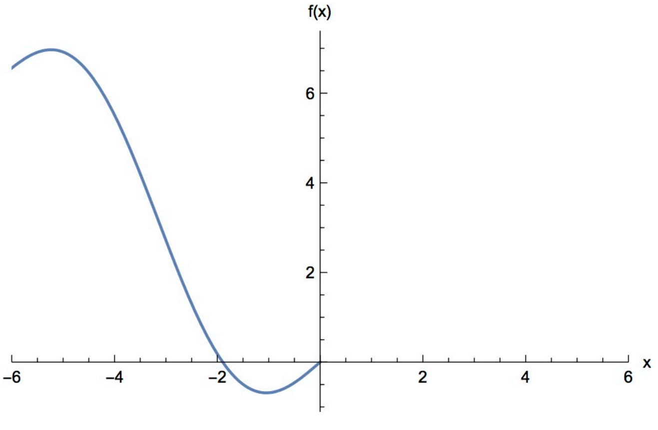
   b) Let $f(x) = x^2$, $g(x) = \sqrt{9+x}$ and $h(x) = (x-1)^{\frac{1}{3}}$. Find:

$$(h \circ ((f \circ g)(g \circ f)))(4)$$

6. Let $f : D_f \to \mathbb{R}$ and $g : D_g \to \mathbb{R}$ be functions given by $f(x) = x^2$ and $g(x) = \sqrt{x}$, respectively. We define the composite $f \circ g : D_{f\circ g} \to \mathbb{R}$, where D_g, D_f and $D_{f\circ g}$ are subsets of $\mathbb{R}$.
   a) What is the domain D_f of f and D_g of g?
   b) What is the range of g?
   Now consider $\boxed{x} \xrightarrow{\mathsf{g}} \boxed{\sqrt{x}} \xrightarrow{\mathsf{f}} \boxed{(\sqrt{x})^2}$
   c) Which values of x do you need to put in the first block so that the expression in the third block defines a real number?
   d) Compute $f(g(x))$ and simplify your expression.
   e) What is the domain of the simplified expression? Let us call it D_*.
   f) Use a) and b) to find the domain $D_{f\circ g}$ of $f \circ g$.
   g) Find the intersection $D_* \cap D_g$. This is the domain of the composite function.
   h) Compare your answers to c, e, f and g. Note that if you had simplified the function in the beginning, you would have come up with the wrong domain for the composite. You have to be careful that at each stage of the map, the numbers that are passed in are within the domain of that part of the composite function.
7. You are given that one of the functions in the table below is linear and the other is exponential. Which is which? Find a formula for each of them.

x	$f(x)$	$g(x)$
-2	36	36
-1	12	20
0	4	4
1	4/3	-12
2	4/9	-28

8. Sketch the graphs of the rational functions given by:
   a) $f(x) = \frac{x-3}{x^2-25}$
   b) $g(x) = \frac{x^3-7x^2+12x}{x^2-16}$
9. For each of the following statements, say whether it is true or false:
   a) If f and g are polynomial functions, $f \circ g$ is a polynomial function.
   b) If g and $f \circ g$ are linear functions, then f is a linear function.
10. If $\sin(x) = \frac{1}{3}$ and $\sec(y) = \frac{5}{4}$ for $0 < x < \frac{\pi}{2}$ and $0 < y < \frac{\pi}{2}$, then find:
    a) $\cos(x + y)$
    b) $\tan(x - y)$
11. Let $f(x) = \sqrt{x}$, $g(x) = \frac{4}{5-x}$ and $h(x) = x^2$. Find:
$$(h \circ ((h \circ g \circ f) - f))(4)$$
    Find the domain and range of $f \circ g$.
12. For each of the following statements, you must decide whether it is true or false.
    If you think it is true, prove it. Otherwise, produce an example to show that the statement is not always true.
    a) No even function is one-to-one.
    b) All odd functions are one-to one.
    c) If f and g are one-to-one functions, then so is their sum $f + g$, their product fg, their difference $f - g$ and their composition $f \circ g$.
    d) If f and g are increasing functions, then, so is their product fg, their sum $f + g$, their difference $f - g$ and their composition $f \circ g$.
13. The following shows part of a graph of a function, f, whose domain is $\mathbb{R}$.



    Sketch the positive half of the graph on the given axes if:
    a) The function is even
    b) The function is odd
    c) For a) and b), draw the graph of $y = |f(x)|$ and $y = f(|x|)$.

Tutorial Questions 5

1. For each of the following equations, write down the set of all real numbers x for which the equation is true.
   a) $\sin(\arcsin x) = x$
   b) $\arcsin(\sin x) = x$
   c) $\arcsin x = \arccos x$
   d) $\ln(x^2) = 2\ln x$
2. Simplify the following expressions:
   a) $\tan(\arcsin(x))$
   b) $\sin(\arctan(x))$
   c) $\sin(2\arccos(x))$
3. Prove that $\sin(\arccos(x)) = \sqrt{1-x^2}$
4. Write the following expressions as a power of e.
   a) 2^x
   b) 3^{4x}
5. Let $f : \mathbb{R} \to \mathbb{R}$ be a function. For each of the following statements say whether it is true or false. Give a reason for your answer.
   a) If f is even, it has an inverse.
   b) If f is odd, it has an inverse.
   c) If f is one-to-one, it is either increasing on $\mathbb{R}$, or it is decreasing on $\mathbb{R}$.
6. Let $f(x) = \dfrac{e^x}{e^{2x}-1}$.
   a) Is f one-to-one? Give a reason for your answer.
   b) Find the inverse of f, if it exists.
7. Claim: "No odd integer can be expressed as a sum of three even integers."
   Prove that this claim is true.
   Hint: Suppose you can express some odd number n as a sum of three even integers then what can you conclude?
8. Find the value of $\cos(\arctan(2) - \arctan(3))$.
9. Write the following expressions as a power of e.
   a) $x^{\sqrt{x}}$
   b) $(\tan(x))^{\sec(x)}$
10. Let $f(x) = e^{\ln(x+1)}$.
    a) Find the domain and range of f.
    b) Draw a rough graph of f.
    c) Does f have an inverse? Give a reason for your answer.
11. Let f be an invertible function with domain $\mathbb{R}$ and inverse f^{-1}.
    a) Prove that if f is an odd function , then f^{-1} is also odd.
    b) Prove that if f is an increasing function, then f^{-1} is also increasing
12. Let f be a function given by $f(x) = \ln(1 - \ln(x^2+\mathrm{e}-1))$
    a) Find the domain of f.
    b) Is f an even function?
    c) Does f have an inverse function? (why?)

Tutorial Questions 6

1. Let f and g be functions given by $f(x) = x^2 + 1$ and $g(x) = \sqrt{x-1}$, respectively. Is $f \circ g$ an odd function?
2. Statement: An integer n is divisible by 3 if and only if n^2 is divisible by 3.

   > *proof:*
   > ($\Longrightarrow$) We first prove that if n is divisible by 3 then n^2 is divisible by 3.
   > Now, if n is divisible by 3 then $n = 3k$ for some integer k, thus $n^2 = 9k^2 = 3(3k^2)$. Notice that $3k^2$ is an integer and so n^2 is divisible by 3.
   > ($\Longleftarrow$) Conversely, we prove that if n^2 is divisible by 3 then n is divisible by 3.
   > This is equivalent to proving that if n is not divisible by 3 then n^2 is not divisible by 3.
   > Now if n is not divisible by 3 then $n = 3k+1$ or $n = 3k'+2$ for some integers k and k', in any case $n^2 = 9k^2+6k+1 = 3(3k^2+2k)+1$ or $n^2 = 9k'^2+12k'+4 = 3(3k'^2+4k'+1)+1$ and thus n^2 is not divisible by 3.$\square$

   a) What does "if and only if" mean?
   b) Is it true that if n is not divisible by 3 then $n = 3k+1$ or $n = 3k+2$ for some integers k and k'? (Why?)
   c) Why is the statement "if n^2 is divisible by 3 then n is divisible by 3" equivalent to statement "if n is not divisible by 3 then n^2 is not divisible by 3."?
   d) Let A and B be statements, is it true in general that $A \implies B \iff \neg B \implies \neg A$?
   ($\implies$ is the logical symbol for implication and $\neg$ a logical symbol for negation.
   One reads the above as "A implies B is equivalent to (not B) implies (not A)".)

3. Use proof by contradiction to prove each of the following statements:
   a) There is no smallest positive real number.
   b) The sum of a rational number and an irrational number is irrational.
   c) $\sqrt{3}$ is irrational. (*Hint:* You will need to use the result in question 2.)
   d) $\sqrt{6}$ is irrational.
4. Use mathematical induction to prove that each of the following statements is true:
   a) For every positive integer n,
   $$1 + 3 + 5 + \cdots + (2n-1) = n^2.$$
   b) For every positive integer n,
   $$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$
   c) For every positive integer n,
   $$2^{2n} - 1$$
   is divisible by 3.
   d) For every positive integer n,
   $$n^3 - n$$
   is divisible by 6.
5. Prove that an integer n is divisible by 5 if and only if n^2 is divisible by 5.
6. Use the above result and a proof technique you have learnt to show that $\sqrt{5}$ is irrational.

7. Read and make sure that you understand the following proof:

> Statement: For $n > 9$ we have that $n^3 > 3n^2 + 3n + 1$.
> *proof:* For $n > 9$, we have
>
> $$\begin{aligned} n^3 &= n \cdot n^2 \\ &> 9n^2 \\ &= 3n^2 + 3n^2 + 3n^2 \\ &= 3n^2 + 3n \cdot n + 3n \cdot n \\ &> 3n^2 + 3n(9) + 3(9)(9) \\ &> 3n^2 + 3n + 1 \square \end{aligned}$$

8. Use a proof technique you have learnt to show that for every integer $n > 9$, we have $2^n > n^3$. One way of doing this uses the fact that for an integer $k \geq 10$:

$$\frac{(k+1)^3}{k^3} = (1 + \frac{1}{k})^3 \leq (11/10)^3 < 2$$

9. Find the flaw in the following proof:

Statement: All cars are the same colour.

Proof: We shall prove this using mathematical induction. To be specific, we shall prove that, for every positive integer n, any set of n cars are the same colour.

Base Case: When $n = 1$, there is only one car in the set, so all the cars in the set have the same colour. This proves the Base Case.

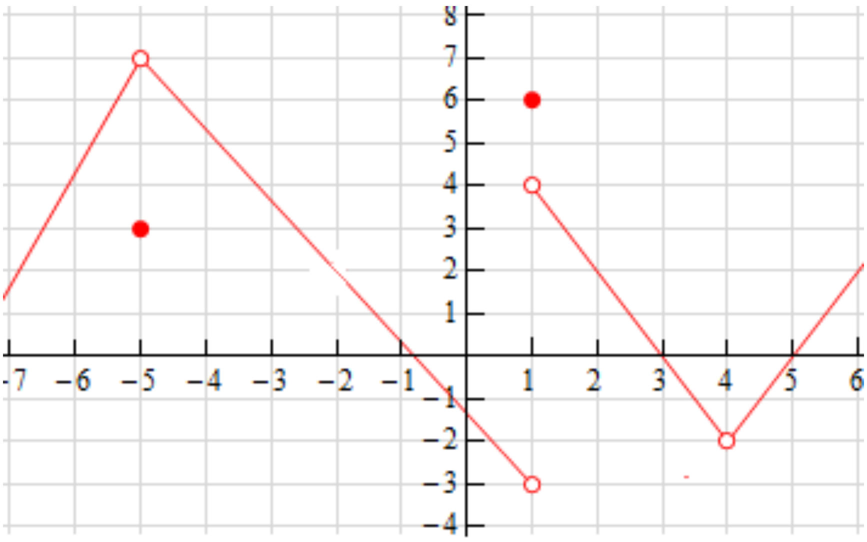
Inductive Step: Suppose that the statement is true for some integer $n \geq 1$. Then any set of n cars are the same colour (this is the Inductive Hypothesis). We must use this to prove that the statement is true for the integer $n+1$. Let S be a set of $n+1$ cars. If x is a car in S, then the set $S - \{x\}$ contains n cars, all of which, by the Inductive Hypothesis, have the same colour. Let's call this colour, whatever it is, 'red'. Let y be a car in S that is different to x. Once again, by the Inductive Hypothesis, all the cars in $S - \{y\}$ are the same colour. But since the colours of the cars in S haven't changed, this means that all the cars in $S - \{y\}$ must also be red. But x is in $S - \{y\}$, which means x is red. Hence, all the cars in S have the same colour.

We have proved that the statement is true for $n = 1$ (the Base Case) and we have proved that whenever the statement is true for an integer n, it is also true for the integer $n + 1$ (the Inductive Step). Hence, by the Principle of Mathematical Induction, the statement is true for all nonnegative integers n.

Tutorial Questions 7

A major hint for some questions here is to use the equation for the difference of two cubes, that is $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$.

1. Quickly find the following limits:
   a) $\lim_{x\to 2}(x^2 - 4x + 1)$
   b) $\lim_{x\to 2} 2^{x+1}$
2. Consider the diagram below as a graph of some function f.



   a) What is $f(-5)$?
   b) Find $\lim_{x\to -5} f(x)$, if it exists.
   c) Find $\lim_{x\to 1^+} f(x)$
   d) Does $\lim_{x\to 1} f(x)$ exist? (Why?)
   e) Is function f defined at $x = 4$?. Does the $\lim_{x\to 4} f(x)$ exist? (Why?)
3. Find the following limits:
   a) $\lim_{x\to\pi} x^2\cos(x)$
   b) $\lim_{x\to 0}\dfrac{\ln(e^x)}{x}$
   c) $\lim_{x\to 16}\dfrac{4-\sqrt{x}}{16x - x^2}$
   d) $\lim_{h\to 0}\dfrac{\sqrt{9+h}-3}{h}$
   e) $\lim_{t\to -2}\dfrac{t^4 - 2}{2t^2 - 3t + 2}$
4. Find the following limits, if they exist. If a limit does not exist, say why.
   a) $\lim_{x\to 2}\dfrac{x^2 - 4}{x^3 - 8}$

b) $\lim_{x\to 0}\left(\frac{2x^2-3x+4}{x}+\frac{5x-4}{x}\right)$

c) $\lim_{x\to 1}\frac{x^3-1}{\sqrt{x}-1}$

d) $\lim_{x\to 1}\frac{x^4-1}{|x^2-2|}$

e) $\lim_{x\to 0^-}\frac{|x|}{2x}$

f) $\lim_{x\to 0^+}\frac{\sqrt{x^3}}{\sqrt{x}}$

Squeeze theorem: If $f(x)\le g(x)\le h(x)$ when x is near a (except possibly at a) and

$$\lim_{x\to a} f(x) = L = \lim_{x\to a} h(x),$$

That is, both limits exist and equal to L.
Then $\lim_{x\to a} g(x) = L$

Example 11 in Stewart 8E, page 69.

Show that $\lim_{x\to 0} x^2\sin\left(\frac{1}{x}\right)=0$

Notice that we cannot use the product law of limits in this case, that is $\lim_{x\to 0} x^2\sin\left(\frac{1}{x}\right)=\lim_{x\to 0}x^2\cdot\lim_{x\to 0}\sin\left(\frac{1}{x}\right)$ since $\lim_{x\to 0}\sin\left(\frac{1}{x}\right)=0$ does not exist.

The fact that $\sin\left(\frac{1}{x}\right)$ is bounded will help us to use Squeeze theorem.
That is $-1\le\sin\left(\frac{1}{x}\right)\le 1$, and multiplying out by x^2
we get $-x^2\le x^2\sin\left(\frac{1}{x}\right)\le x^2$
Notice that the fact that $x^2\ge 0$ is used here. (This is important, make sure you know why if not ask a tutor).

Now notice that $\lim_{x\to 0} -x^2=0$ and $\lim_{x\to 0} x^2=0$

Therefore by Squeeze theorem we have $\lim_{x\to 0} x^2\sin\left(\frac{1}{x}\right)=0$.

5. Use Squeeze theorem to prove that $\lim_{x\to 0} x^4\cos\left(\frac{\pi}{x}\right)=0$
6. Find $\lim_{x\to 0} x^2\arctan\left(\frac{1}{x^4}\right)$ (Hint: Use Squeeze theorem)
7. Find the following limits:

a) $\lim_{x\to 0}\frac{\sqrt[3]{x+27}-\sqrt[3]{2x+27}}{x}$

b) $\lim_{x\to 0}\left[x^3\arctan\left(\frac{1}{x}\right)\right]$

c) $\lim_{x\to 0}\frac{(3+x)^3-27}{x}$

d) $\lim_{x\to 0}\frac{\sqrt{x^2+16}-4}{x^2}$

e) $\lim_{x\to 0} f(x)$ where $f(x) = \begin{cases} x^2, & \text{if } x \text{ is a rational number} \\ x, & \text{if } x \text{ is an irrational number} \end{cases}$

8. For each of the following statements, say whether it is true or false. If true, prove it; if false, give a counter-example.
   a) If $\lim_{x\to a}[f(x) + g(x)]$, then both $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist.
   b) If $\lim_{x\to a}[f(x) + g(x)]$ and $\lim_{x\to a} f(x)$ exist, then $\lim_{x\to a} g(x)$ exists.
   c) If $\lim_{x\to a} f(x)$ exists and $\lim_{x\to a} \frac{f(x)}{g(x)} = 0$, then $\lim_{x\to a} g(x)$ exists.
   d) If $\lim_{x\to a}[f(x)]^2$ exists, then $\lim_{x\to a} f(x)$ exists.

Recall: A function f is continuous at a if $\lim_{x\to a} f(x) = f(a)$.
We will learn more about continuity next term.

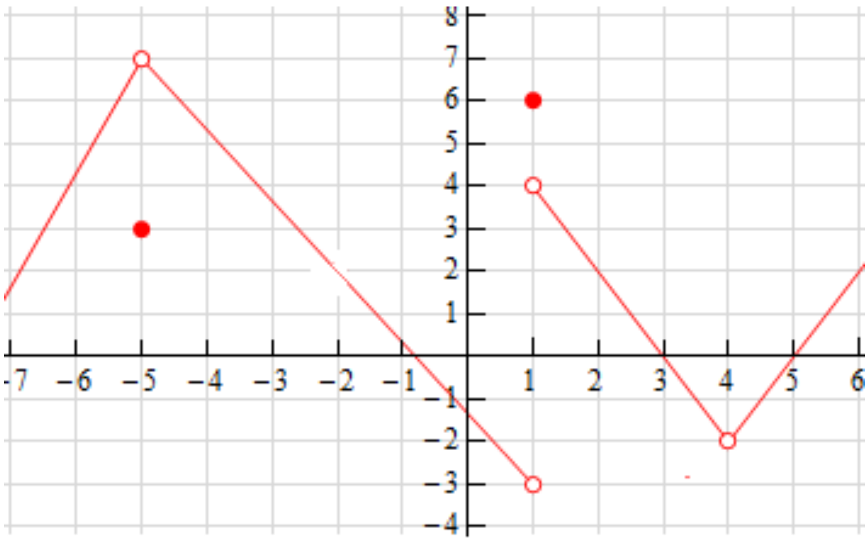
9. If you look at the graph of the function $f(x) = \sqrt{x}$, it seems as if this function is continuous on $[0, \infty)$. In this question you will prove that this is the case, using the Squeeze (or Sandwich) Theorem.
   a) Show that for $a > 0, a + h > 0$,
   $$|\sqrt{a+h} - \sqrt{a}| = \frac{|h|}{\sqrt{a+h} + \sqrt{a}} < \frac{|h|}{\sqrt{a}}.$$
   [*Hint:* Multiply by $\frac{\sqrt{a+h} + \sqrt{a}}{\sqrt{a+h} + \sqrt{a}}$.]
   b) Use the Squeeze theorem to deduce that $\lim_{h\to 0} \sqrt{a+h} = \sqrt{a}$ for all $a > 0$.
   c) Now show that $\lim_{h\to 0^+} \sqrt{h} = 0$ (you may assume this limit exists). Hint: try assuming that $\lim_{h\to 0^+} \sqrt{h} = c \neq 0$ and obtaining a proof by contradiction.
   d) Explain why it follows from (b) and (c) that f is continuous on $[0, \infty)$.
10. Bonus question: The length of the edge of the Koch Snowflake.

Tutorial Questions 8

1. Evaluate the limit and justify each step by indicating the appropriate limit properties.
   a) $\lim_{x\to\infty} \frac{5x^2-3x+5}{2x^2+3x+1}$

   b) $\lim_{x\to\infty} \sqrt{\frac{32x^3+5x+1}{3+4x^2+2x^3}}$

2. Consider the diagram below as a graph of some function f.



   a) Is the above function continuous at 3? (Why?)
   b) Give three values for which f fails to be continuous and give a reason why it is not continuous at each of those values.
   (Make sure you understand the definition of continuity).
   c) In one of the values, the discontinuity can be removed by enlarging the domain of f. How can this be done?

3. Evaluate the following limits:
   a) $\lim_{x\to\infty} \frac{\sqrt{x^2-3x+5}}{x+2}$

   b) $\lim_{x\to-\infty} \frac{\sqrt{x^2-3x+5}}{x+2}$

4. Use the definition of continuity and properties of limits to show that:
   a) $f(x) = \frac{x^2+5x}{2x-1}$ is continuous at $x = 1$.
   b) $g(x) = \frac{x^2}{\sqrt{x+2}}$ is continuous at $x = 2$.

5. Let $f(x) = \dfrac{1}{3x-1}$.
   a) Without finding the derivative of f, say whether you expect $f'(1)$ to be positive or negative. Give a reason for your answer.
   b) Use only the **definition** of the derivative to find $f'(1)$.
   c) Find the equation of the tangent line to the graph of f at the point where $x = 1$.

6. Find the equation of the tangent line to the curve at the given point.
   a) $y = \sqrt{x}$, point $(16, 4)$.

b) $y = \frac{2x+1}{x+2}$, point $(1, 1)$.

7. Find the limit or show that it does not exist.

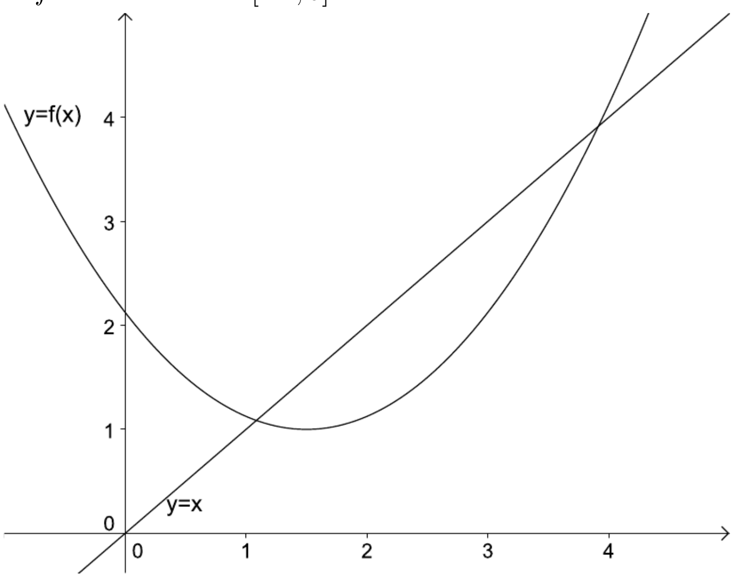
   a) $\lim\limits_{t\to\infty} \frac{\sqrt{t}+t^2}{2t-t^2}$

   b) $\lim\limits_{x\to\infty} \frac{\sqrt{x^2-3x+5}}{x+2}$

   c) $\lim\limits_{x\to\infty} \left(\sqrt{9x^2+x} - 3x\right)$

   d) $\lim\limits_{x\to\infty} \frac{x^2}{\sqrt{x^4+1}}$

8. The graph of the function f on the interval $[-1, 5]$ is shown below.



   Arrange the following numbers in increasing order, and explain your answer:
   (a) 0 (b) 1 (c) $f'(1)$ (d) $f'(2)$ (e) $f'(4)$ (f) $\frac{1}{2}(f(4) - f(2))$

9. Use the intermediate Value Theorem (IVT) to show that the following functions have a zero in the specified interval.

   a) $f(x) = x^4 + x - 3$, interval $(1, 2)$.

   b) $g(x) = \sqrt[3]{x} + x - 1$, interval $(0, 1)$.

   c) $h(x) = \sqrt{x} + \frac{2}{x} - x$, interval $(2, 3)$.

10. Use the Squeeze theorem to evaluate $\lim\limits_{x\to\infty} \frac{\sin(x)}{x}$

11. Suppose f and g are continuous functions such that $g(2) = 6$ and $\lim\limits_{x\to 2}[3f(x) + f(x)g(x)] = 36$. Find $f(2)$.

12. Evaluate the following limits.

    a) $\lim\limits_{x\to\infty} \left(\sqrt{x^2+ax} - \sqrt{x^2+bx}\right)$

    b) $\lim\limits_{t\to\infty} \frac{t-t\sqrt{t}}{2t^{\frac{3}{2}}+3t-5}$

13. Find all values of a such that f is continuous on $(-\infty, \infty)$

$$f(x) = \begin{cases} x+1 & \text{if } x \le a \\ x^2 & \text{if } x > a \end{cases}$$

14. Is there are a real number satisfying equation $2^x = x^5 + 4$? (hint: define $f(x) = 2^x - x^5 - 4$.)

15. For each of the following, say whether it is true or false. If true, prove it; if false, give a counter-example. **(Note: Giving an example does not count as a proof for a true statement).**
    a) If f is continuous at a, then f is differentiable at a.
    b) If f is differentiable at a, then f is continuous at a.
16. Determine whether $f'(0)$ exists.
    a)
    $$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$
    b)
    $$g(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$
17. Find the values of a and b that make function f continuous everywhere on $(-\infty, \infty)$
    $$f(x) = \begin{cases} \frac{x^2-4}{x-2} & \text{if } x < 2 \\ ax^2 - bx + 3 & \text{if } 2 \leq x < 3 \\ 2x - a + b & \text{if } x \geq 3 \end{cases}$$
18. The function f is defined by
    $$f(x) = \begin{cases} x^3 & \text{for } x < 0 \\ x^2 & \text{for } x \geq 0 \end{cases}$$
    a) Sketch the graph of f.
    b) For each of the following statements, say whether it is true or false and prove that your answer is correct:
        i. f is continuous at $x = 0$.
        ii. f is differentiable at $x = 0$ (make sure you use the limit definition of the derivative in your answer).
        iii. f' is continuous at $x = 0$.
        iv. f' is differentiable at $x = 0$ (make sure you use the limit definition of the derivative in your answer).

Tutorial Questions 9

1. You are given that the function $y = f(x)$ has the graph below.

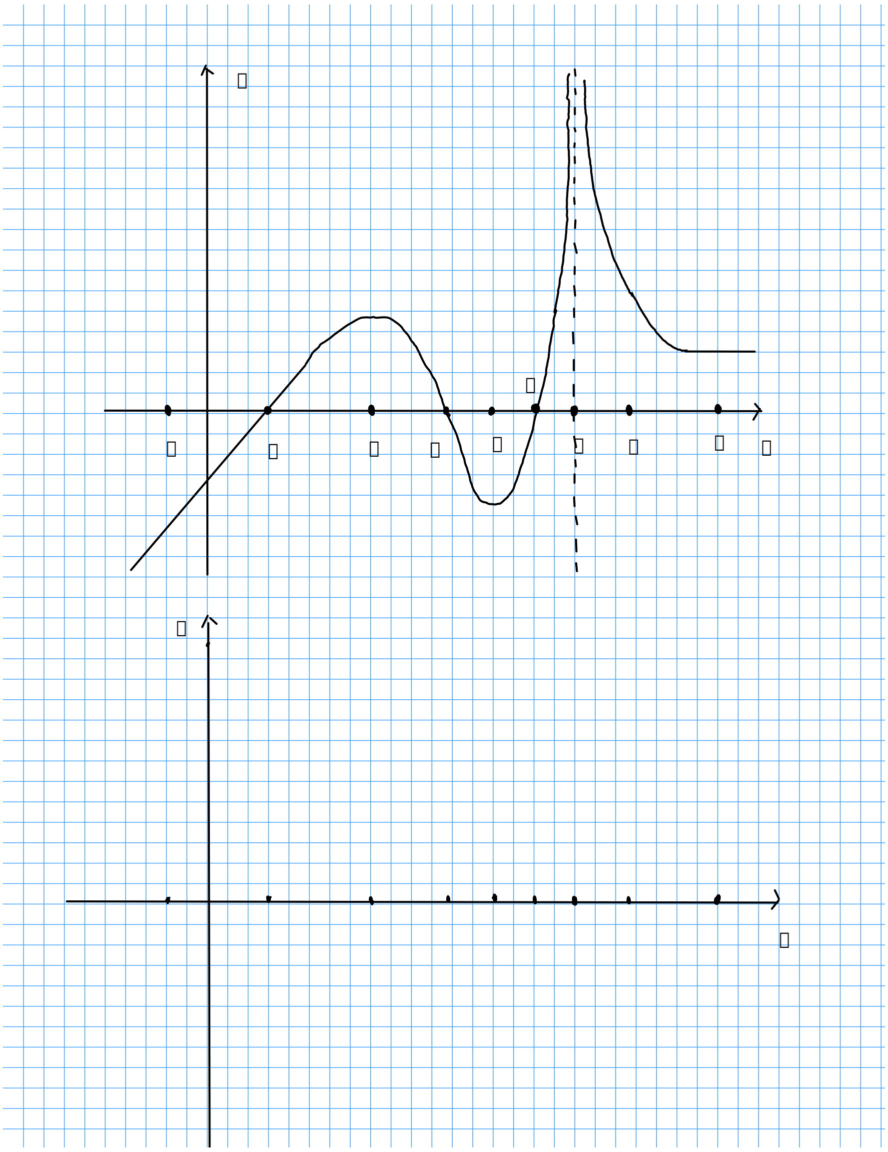


Figure 0.1: The function $y = f(x)$

On the given axes, sketch the graph of $y = f'(x)$ wherever it is defined (do not worry about the values).
[*Hint:* Don't try to guess a formula for $f(x)$; there isn't one! At each of the indicated points, ask yourself more-or-less what $f'(x)$ is at that point.]

2. Let f be a function with domain $\mathbb{R}$. For each of the following statements say whether it is true or false:
   a) If $f''(x) = 0$ for all $x \in \mathbb{R}$, then f is a linear function.
   b) If $f''(x) > 0$ for all $x \in \mathbb{R}$, then $f'(x) > 0$ for all $x \in \mathbb{R}$.
   c) If $f'(x) > 0$ for all $x \in \mathbb{R}$, then $f''(x) > 0$ for all $x \in \mathbb{R}$.

   Now give reasons for all your answers. *Remember:* If you think the statement is false, it is good enough to give an example of a function for which the statement is false. If you think it it is true, you have to give a reason or proof.

3. Let f and g be differentiable functions, then for each of the following say whether it is true or false— prove it if true or give a counter-example if false.
   a) $(fg)'(x) = f'(x)g'(x)$.
   b) $\left(\frac{f}{g}\right)'(x) = \frac{f'(x)}{g'(x)}$ for $g(x) \neq 0$.
4. Differentiate the following functions:
   a) $f(x) = \sqrt[3]{x}(4 - x)$
   b) $g(x) = \frac{x^2+4x+1}{\sqrt{x}-1}$
   c) $h(x) = \left(\frac{1}{x} - \frac{1}{\sqrt{x}}\right)^2$
   d) $k(x) = \frac{(x+2)^9}{1-x}$

5. If $f(x) = \sqrt{x} \cdot g(x)$, where $g(4) = 8$ and $g'(4) = 5$, then find $f'(4)$.
6. You are given the following information:

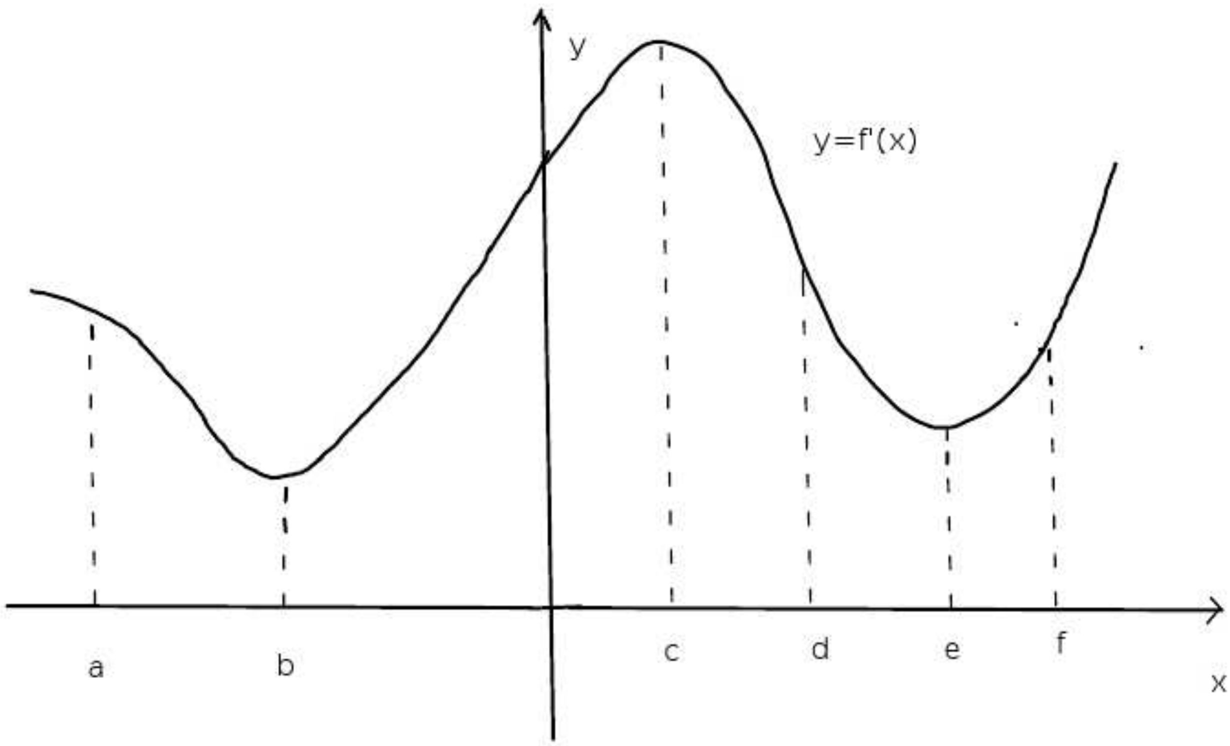
x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
-1	1	-1	-2	2
1	-2	1	3	-1

   Find the following:

   (a) $\left.\dfrac{d}{dx}[f(x)g(x)]\right|_{x=-1}$ (b) $\left.\dfrac{d}{dx}\left[\dfrac{g(x)}{f(x)}\right]\right|_{x=1}$ (c) $\left.\dfrac{d}{dx}[f(g(x))]\right|_{x=1}$

   (d) $\left.\dfrac{d}{dx}[g(f(x))]\right|_{x=-1}$ (e) $\left.\dfrac{d}{dx}[f(f(x))]\right|_{x=-1}$ (f) $\left.\dfrac{d}{dx}[f(x^{\pi+1})]\right|_{x=1}$

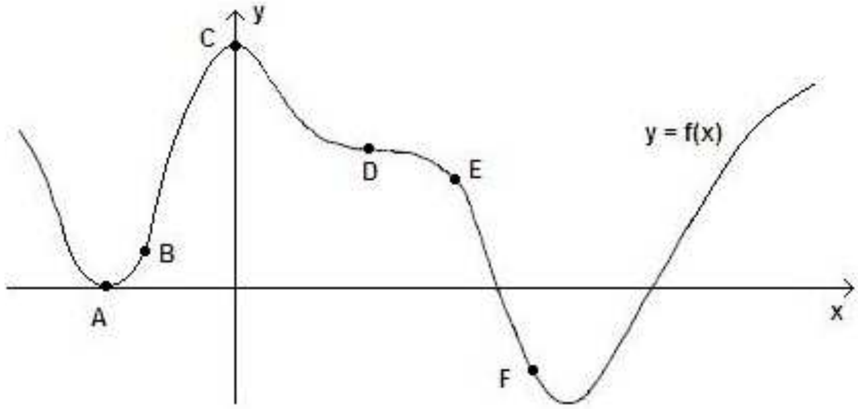
   (g) $\left.\dfrac{d}{dx}[3(f(x))^2 - 5g(x^2)]\right|_{x=1}$
7. Show from the definition of the derivative that the derivative of the function f given by $f(x) = \cos(x)$ is $f'(x) = -\sin(x)$. (Hint: You may use the facts that $\lim_{x\to 0} \frac{\sin(x)}{x} = 1$ and $\lim_{x\to 0} \frac{1-\cos(x)}{x} = 0$)
8. The graph of the *derivative* f' of a function f is given below.



At which of the marked points is

(a) $f(x)$ the largest? (b) $f(x)$ the smallest? (c) $f'(x)$ the largest?
(d) $f'(x)$ the smallest? (e) $f''(x)$ the largest? (f) $f''(x)$ the smallest?

9. Evaluate the following limits:
   a) $\lim_{x\to 0} \frac{\sin(5x)}{8x}$
   b) $\lim_{x\to 0} \frac{\sin(2x)\sin(3x)}{x^2}$
   c) $\lim_{x\to 0} \frac{\cos(x)-1}{\sin(x)}$
   d) $\lim_{x\to 0} \frac{\sin(x^2)}{x}$
10. At which of the marked points on the graph of the function f given below are
    a) f' and f'' both not zero and have the same sign?
    b) at least two of f, f' and f'' equal to 0?



11. Evaluate the following limits:
   a) $\lim\limits_{x\to 0} \frac{\sin(x)}{x+\tan(x)}$
   b) $\lim\limits_{x\to 1} \frac{\sin(x-1)}{x^2+x-2}$ [*Hint:* Let $u = x - 1$]
   c) $\lim\limits_{x\to \frac{\pi}{4}} \frac{1-\tan(x)}{\sin(x)-\cos(x)}$
   d) $\lim\limits_{x\to 0} \csc(x)\sin(\sin(x))$
12. Differentiate the following functions:
   a) $f(x) = x^4 \cdot \sin(2x) \cdot \tan(x)$
   b) $g(x) = \frac{\sec^2(x)}{1+\tan(x)}$
   c) $h(x) = \tan\left(\sqrt[3]{\cos(x)}\right)$
13. In class, we proved the quotient rule using the definition of a derivative. Use the product rule and chain rule to prove the quotient rule. $\left(\text{hint: } \frac{f(x)}{g(x)} = f(x) \cdot \frac{1}{g(x)}\right)$, where $g(x) \neq 0$.
14. Suppose f is a one-to-one function with $f(0) = 3$ and $f'(x) = \sqrt{(f(x))^2 - 1}$, the find $(f^{-1})'(3)$. (hint: $(f \circ f^{-1})(x) = x$).
15. If $F(x) = f(xf(xf(x)))$, where $f(1) = 2$, $f(2) = 3$, $f'(1) = 4$, $f'(2) = 5$ and $f'(3) = 6$, then find $F'(1)$. (hint: Use chain rule and product rule carefully).
16. (Important: Comprehension question)
   The following theorem gives a different way of thinking about differentiability. You have seen in lectures a definition to say what it means for a function to be differentiable at a point. The theorem below says that we could have given another definition, and that these two definitions are *equivalent*.
   Theorem. *The function f is differentiable at $x = a$ if and only if there is a constant m and a function E of x, defined for all $x \neq a$, such that*
   - $f(x) = f(a) + m(x - a) + E(x)(x - a)$ *for all $x \neq a$, and,*
   - $\lim\limits_{x\to a} E(x) = 0$.

   (If both these conditions are satisfied, then $f'(a) = m$.)
   Proof. If f is differentiable at $x = a$, then $f'(a) = \lim\limits_{x\to a} \dfrac{f(x) - f(a)}{x - a}$ exists.
   For $x \neq a$, put $E(x) = \dfrac{f(x) - f(a)}{x - a} - f'(a)$. Then
$$\lim_{x\to a} E(x) = \lim_{x\to a} \left[\frac{f(x) - f(a)}{x - a} - f'(a)\right] = f'(a) - f'(a) = 0, \qquad (1)$$

and

$$E(x)(x-a) = f(x) - f(a) - f'(a)(x-a). \qquad (2)$$

If we put $m = f'(a)$, we have

$$f(x) = f(a) + m(x-a) + E(x)(x-a).$$

This proves the first half of the theorem. (3)

For the second half, suppose there is a constant m and a function E of x such that $f(x) = f(a) + m(x-a) + E(x)(x-a)$ and $\lim_{x\to a} E(x) = 0$. Then

$$\lim_{x\to a} \frac{f(x)-f(a)}{x-a} = \lim_{x\to a}[m + E(x)] = m. \qquad (4)$$

Hence f is differentiable at $x = a$. (5) ■

a) You may be used to the definition which says f is differentiable at $x = a$ when $\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ exists. Is this the same as saying $\lim_{x\to a} \frac{f(x)-f(a)}{x-a}$ exists? Give a reason for your answer.

b) Explain what we use in line (1) to say that $\lim_{x\to a} \left[\frac{f(x)-f(a)}{x-a} - f'(a)\right] = f'(a) - f'(a)$.

c) The function E has been defined for $x \neq a$. If we want to define E for $x = a$ as well, in such a way that E is continuous at $x = a$, what should $E(a)$ be?

d) How is the equation in line (2) obtained?

e) What is the *first half* of the theorem mentioned in line (3), i.e., is it the “if” or the “only if” part of the proof? Why can we say it has been proved by the time we get to line (3)?

f) Why can we say in line (4) that $\lim_{x\to a} \frac{f(x)-f(a)}{x-a} = \lim_{x\to a}[m + E(x)]$?

g) How does it follow, also in line (4), that $\lim_{x\to a}[m + E(x)] = m$?

h) Why can we say in line (5) that f is differentiable at $x = a$?

i) We could say that this theorem says that a function f is differentiable at a if and only if it can be approximated by a linear function near a. Why can we say this?

Tutorial Questions 10

1. For each of the following statements say whether it is true or false, and give a reason for your answer:
   a) The rate at which the function $h(x) = e^{x^3}$ changes at $x = -1$ is 3 times as large as the rate at which $g(x) = e^x$ changes at $x = -1$.
   b) If f is differentiable at $x = 2$, then the rate at which the function $g(x) = x^2 f(x)$ changes at $x = 2$ is 4 times as large as the rate at which f changes at $x = 2$.
2. For each of the following, find $\frac{dy}{dx}$.
   a) $x^2 + y^3 = 2018$
   b) $\cos(x) + \sin(y) = e^y$
   c) $y = \sqrt{\arctan(x)}$
3. Write the following expressions as a power of e.
   a) $(\sqrt{x})^{3x}$
   b) $\sec(x)^{\ln(x^2+1)}$
4. Differentiate the functions given by the following:
   a) $f(x) = x\ln(x) - x$
   b) $g(x) = \cos(\ln(x))$
   c) $h(x) = \log_2(1 + \tan(x)e^x)$
   d) $k(x) = x^{3x}$
5. Use logarithmic differentiation to find the derivatives of the following:
   a) $y = (\sqrt{x})^x$
   b) $y = \sqrt{\frac{x-1}{x^4+1}}$
   c) $y = \sqrt{x}e^{x^2-x}(x+1)^{\frac{2}{3}}$

6. Let $f(x) = x\arctan\left(\frac{1}{x}\right)$ if $x \neq 0$ and $f(0) = 0$.
   a) Is f continuous at 0?
   b) Is f differentiable at 0?

7. Find $\frac{dy}{dx}$ for each of the following:
   a) $\arctan(y) + x^4 = \arcsin(x) + 2y$
   b) $y = \arccos(\sqrt{x})$
   c) $y = \cot^{-1}(x) + \cot^{-1}\left(\frac{1}{x}\right)$
   d) $y = \arcsin(e^x) + xy^2$
   e) $y = \sqrt{1-x^2}\arcsin(x)$
8. Let $f(x) = cx + \ln(\cos(x))$. Find the value of c such that $f'(\frac{\pi}{4}) = 6$.
9. Find $\frac{d^9}{dx^9}(x^8\ln(x))$

10. Let n be a positive integer, then find a formula for $\frac{d^n y}{dx^n}$ if $y = \ln(x-1)$.
    (hint: find the first few derivatives and make an intelligent guess, then use induction to prove your guess.)
11. For each of the following, find $\frac{dy}{dx}$.
    a) $y = \log_2(x\log_5 x)$
    b) $3^{\cos(2x)} + \ln(y^2+1) = \tan(xy)$
    c) $y = \ln(\ln(\ln(x)))$

d) $y^2 = \arctan\left(\frac{x}{a}\right) + \ln\left(\sqrt{\frac{x-a}{x+a}}\right)$

e) $y = \arccos\left(\frac{b+a\cos(x)}{a+b\cos(x)}\right)$, where $0 \leq x \leq \pi$, $a > b > 0$

f) Real challenge: $y = \left(\sin\left(e^{\cos\left(\frac{1}{x}\right)}\right)\right)^{\frac{1}{x^2}}$

12. Prove that for $x > 0$, $\arctan(x) + \arctan(\frac{1}{x}) = \frac{\pi}{2}$ (hint: you may use the fact that if $f'(x) = 0$ for all x in interval I, then for all $x \in I$, $f(x) = C$ for some $C \in \mathbb{R}$)

13. The equation

$$2(x^2 + y^2)^2 = 25(x^2 - y^2)$$

defines a curve in the xy-plane.

a) Verify that 0 is the only y-intercept of the curve.

b) Find the x-intercepts of the curve.

c) Is the curve symmetric about the x-axis? And about the y axis? Why?

d) Show that the point $(3, 1)$ lies on the curve. Find three other points on the curve (other than the intercepts with the axes).

e) Is the curve the graph of a function? Why?

f) Find $\dfrac{dy}{dx}$ by differentiating implicitly.

g) Find the slope of the tangent line at the point $(3, 1)$, and an equation of the tangent line to the curve at $(3, 1)$.

h) Find all the points on the curve where the tangent line is horizontal.

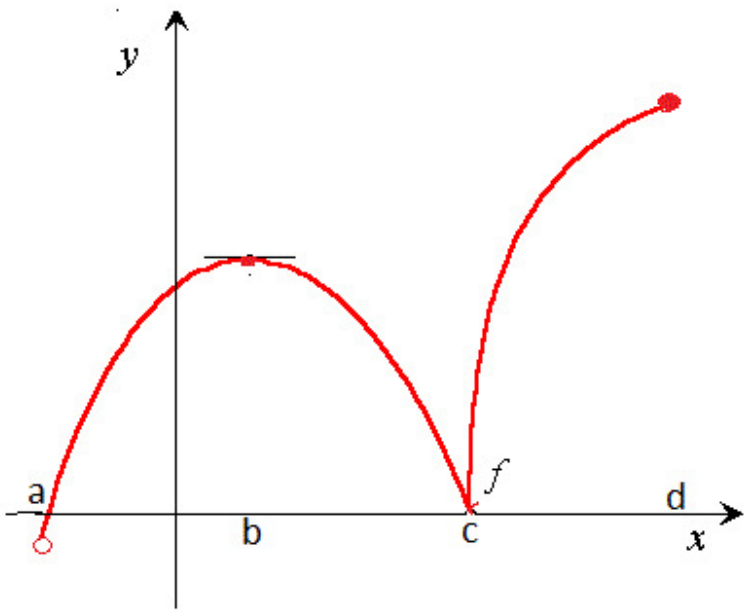
i) Find the points on the curve where the tangent line is vertical.

j) *(A bit more of a challenge)* It is quite difficult to see from the information we have so far what the curve will look like. Here is a way of using a change of variables to get a better idea of the shape of the curve. Make the substitution $x = r\cos\theta, y = r\sin\theta$, where $r \geq 0$ and $0 \leq \theta < 2\pi$. Draw a sketch to see how x, y, r and θ are related. (The r and θ are known as the *polar coordinates* of the point (x, y); the polar coordinates of a point are unique, except for the origin, where we take $r = 0$ and allow any θ.) Make these substitutions in the equation $(*)$ to obtain an equation in r and θ, and simplify it as far as possible using trig identities. Now try to trace the curve by starting with $\theta = 0$ and increasing it, and finding the corresponding r. (Remember that a square can never be negative!) Make a sketch of the curve.

Tutorial Questions 11

Drawing a diagram for related rate problems is advisable as this makes the problem easier than it looks when written in word form.

1. Consider the diagram below as a graph of some function f.



   a) Is f differentiable everywhere on the open interval (a, d)?
   b) Locate the critical points of f.
   c) Does f have an absolute or global maximum? Does f have an absolute or global minimum? (why?)
2. For each of the following, say whether it is true or false.
   a) If f has a local minimum, then f has an absolute minimum.
   b) If f has an absolute maximum, then f has a local maximum.
3. Prove the following:
   a) $\frac{d}{dx}\cot^{-1}(x) = -\frac{1}{1+x^2}$
   b) $\frac{d}{dx}\csc^{-1}(x) = -\frac{1}{x\sqrt{x^2-1}}$
   c) $\frac{d}{dx}\sec^{-1}(x) = \frac{1}{x\sqrt{x^2-1}}$
4. Prove that $\arcsin(x) + \arccos(x) = \frac{\pi}{2}$.
5. If $x^2 + y^2 + z^2 = 9$, $\frac{dx}{dt} = 5$, $\frac{dy}{dt} = 4$, then find $\frac{dz}{dt}$ when $x = 2, y = 2$ and $z = 1$
6. Each side of a square is increasing at a rate of 6 cm/s. At what rate is the area of the square increasing when the area of the square is 16 cm^2?
7. Two cars start moving from the same point. One travels south at 30 km/h while the other travels west at 72 km/h. At what rate is the distance between the cars increasing 4 hours later.
8. For each of the following, Find the absolute minimum and maximum values.

a) $f(x) = x^3 - 3x^2 + 1$ on the closed interval $[-\frac{1}{2}, 4]$.
b) $g(x) = x - \sqrt[3]{x}$ on the closed interval $[-1, 4]$.
c) $h(x) = x + \cot(\frac{x}{2})$ on the closed interval $[\frac{\pi}{4}, \frac{7\pi}{4}]$.
d) $k(x) = \frac{\sqrt{x}}{1+x^2}$ on the closed interval $[0, 2]$.

9. For each of the following, say whether it is true or false. Prove it if true and give a counter-example if false.
   a) If f is differentiable at a, then f'' at a exists.
   b) If $f'(a)$ exists and $f'(a) = 0$, then f has a local minimum or local maximum at a.
   c) Suppose f has a local maximum at a, then $f'(a)$ exists and $f'(a) = 0$.
10. Prove the following identity
$$\arcsin\left(\frac{x-1}{x+1}\right) = 2\arctan(\sqrt{x}) - \frac{\pi}{2}$$
.
11. A particle moves along the curve $y = 2\sin(\frac{\pi x}{2})$. As the particle passes through the point $(\frac{1}{3}, 1)$, its x-coordinate increases at a rate of $\sqrt{10}$ cm/s. How fast is the distance from the particle to the origin changing at this instance?
12. **These are from an older version of Stewart. Similar problems and how to solve them can be found on pages 185-189 of Stewart Calculus 9E.**
   a) A light shines from the ground up towards a wall $10m$ away. If a man $1.8m$ tall walks from the spotlight toward the building at a speed of $1.4\ ms^{-1}$, how fast is the length of his shadow on the building decreasing when he is $3m$ from the building?
   b) Water is leaking out of an inverted conical tank at a rate of $10,000\frac{cm^3}{min}$ at the same time that water is being pumped into the tank at a constant rate. The tank has height $6m$ and the diameter at the top is $4m$. If the water level is rising at a rate of $20\frac{cm}{min}$ when the height of the water is $2m$, find the rate at which water is being pumped into the tank.
   c) A lighthouse is located on a small island 3 km away from the nearest point P on a straight shoreline and its light makes four revolutions per minute. How fast is the beam of light moving along the shoreline when it is 1 km from P?
   d) A runner sprints around a circular track of radius $100m$ at a constant speed of $7\frac{m}{s}$. The runner's friend is standing at a distance 200 m from the center of the track. How fast is the distance between the friends changing when the distance between them is 200 m?

   I highly recommend completing as many questions in Stewart on related rates as you can. They take time and practice to master.
13. If a and b are positive numbers, then find the maximum value of $f(x) = x^a(1-x)^b$.
14. Prove that $p(x) = x^3 + x - 1$ has *exactly* one real root in interval $[0, 1]$.
15. Prove that the function given by $f(x) = x^{2017} + x^{1995} + x + 1$ has neither a local maximum nor a local minimum. (hint: Prove by contradiction)
16. Show that the equation $x^4 + 4x + c = 0$ has at most two real roots.
17. Verify that $f(x) = x^3 - 2x^2 - 4x + 2$ satisfies the three hypotheses of Rolle's theorem on the interval $[-2, 2]$ and find all numbers c that satisfy the conclusion of Rolle's theorem.
18. Read the statement and proof of the following theorem and then answer the questions following it.

   Theorem: If a function f is differentiable at $x = c$ and $f(c)$ is a local maximum, then $f'(c) = 0$.
   Proof: There is an open interval I which contains c such that
   for all $x \in I$, $f(x) \leq f(c)$. (1)
   It follows that if h is small enough, $f(c+h) - f(c) \leq 0$. (2)
   If $h < 0$, we therefore have $\dfrac{f(c+h) - f(c)}{h} \geq 0$, (3)

and if $h > 0$, $\dfrac{f(c+h) - f(c)}{h} \leq 0$. (4)

From this we can deduce that $\displaystyle\lim_{h \to 0^-} \frac{f(c+h) - f(c)}{h} \geq 0$ (5)

and $\displaystyle\lim_{h \to 0^+} \frac{f(c+h) - f(c)}{h} \leq 0$. (6)

Since f is differentiable at $x = c$, $\displaystyle\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = 0$ (7)

and so $f'(c) = 0$.

a) From which assumption in the statement of the theorem does line (1) in the proof follow?

b) Why does h have to be "small enough" for the inequality in line (2) to follow from line (1)?

c) How do the inequalities in lines (3) and (4) follow from line (2)?

d) State the limit rule we seem to be using in lines (5) and (6).

e) Why can we say, in line (7), that since f is differentiable at $x = c$,
$\displaystyle\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = 0$?

f) The theorem stated here is also true if we replace "local maximum" by "local minimum". This can be proved in the same way, with appropriate changes to the inequality signs. Here is another possibility: Try to prove the theorem for a local minimum by applying the theorem for a local maximum to the function $g(x) = -f(x)$.

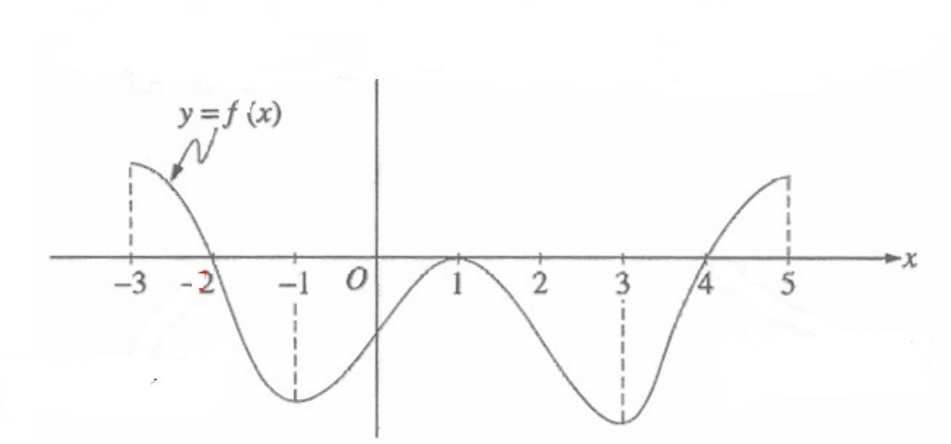
g) Is the following limit theorem true: If $f(x) < k$ for all x near a and $\displaystyle\lim_{x \to a} f(x)$ exists, then $\displaystyle\lim_{x \to a} f(x) < k$? Give a reason for your answer.

h) Is it true that every stationary point is a local maximum or local minimum?

Tutorial Questions 12

If you feel that you need more practice with differentiation then do lots of examples from Stewart 9E— 2.3 Exercises 1-52, 2.4 Exercises 1-22, 2.5 Exercises 1-56, 2.6 Exercises 1-34

1. Consider the diagram below as a graph of some function f.



   a) Find open intervals on which f is increasing. Similarly open intervals on which f is decreasing.
   b) Locate stationary points and specify their nature (local minimum or local maximum).
   c) Find open intervals on which f is concave up. Similarly open intervals on which f is concave down.
   d) Locate or estimate points of inflection on the graph of f.
2. Suppose $f(3) = 2$, $f'(3) = \frac{1}{2}$, and $f'(x) > 0$ and $f''(x) < 0$ for all $x \in \mathbb{R}$.
   a) Sketch a possible graph for f. (Confirm your answer with a tutor.)
   b) How many solutions does the equation $f(x) = 0$ have?
   c) Is it possible that $f'(2) = \frac{1}{3}$? (Why?)
3. Let $f(x) = x^3 e^{-x}$.
   a) Find $f'(x)$ and $f''(x)$.
   b) Find a sign table for f, f' and f''. [This means that you have to draw up a table which indicates for which x each of $f(x), f'(x)$ and $f''(x)$ will be positive, and for which x each of them will be negative. You should also mark the point(s) (if any) where each of them equals 0.]
   c) For which x is f increasing, and for which x is f decreasing?
   d) Where (if anywhere) does the graph of f have a horizontal tangent?
   e) For which x is the graph of f concave up? And concave down?
   f) At which values of x does the concavity of the graph of f change (i.e., what are the points of inflection)?
   g) What is $\lim_{x\to-\infty} f(x)$?
   h) What is $\lim_{x\to\infty} f(x)$? *[Hint:* We can write $f(x) = \dfrac{x^3}{e^x}$. You need to decide which grows faster as $x \to \infty$: x^3 or e^x.)]

i) Draw a rough graph of f using all this information.

4. For each one of the following statements, say whether it is true or false.[Give a **proof** if true, otherwise a **counter-example**].
   a) A polynomial of degree four cannot have exactly one point of inflection.
   b) A point of inflection cannot be a critical point.
   c) A local minimum cannot be a point of inflection.
   d) At a point of inflection of the graph of a function the second derivative is zero.
   e) If $f'(a) = 0 = f''(a)$, then the graph of f has a point of inflection at $x = a$.
   f) If the graph of f has a point of inflection at $x = a$, then $f'(a) = 0 = f''(a)$.
   g) If $f'(a)$ does not exist, then f cannot have a local maximum or minimum at $x = a$.
   h) If $f''(a)$ does not exist, the graph of f cannot have a point of inflection at $x = a$.
5. Let $f(x) = (x-3)^{-2}$. Show that there is no value of c in $(1, 4)$ such that $f(4)-f(1) = f'(c)(4-1)$. Why does this not contradict the Mean Value theorem?
6. Let $f(x) = 2 - |2x - 1|$. Show that there is no value of c such that $f(3) - f(0) = f'(c)(3 - 0)$. Why does this not contradict the Mean Value Theorem?
7. Use l'Hospital's Rule to evaluate the following limits:
   a) $\lim\limits_{x\to 0} \frac{\sin(x^2+x)}{x}$
   b) $\lim\limits_{x\to 0^+} x^x$
   c) $\lim\limits_{x\to\infty} \frac{\ln(x)+2018}{\sqrt[3]{x}}$
   d) $\lim\limits_{x\to 0} \frac{5^x-4^x}{3^x-2^x}$
   e) $\lim\limits_{x\to -\infty} xe^x$
   f) $\lim\limits_{x\to 0^+} x\ln x$
8. Sketch the following curves:
   a) $y = \frac{x^2}{x^2+9}$
   b) $y = (x - 3)\sqrt{x}$
   c) $y = \sqrt{x^2 + x - 2}$
   d) $y = 4 + \frac{1}{x} + \frac{1}{x^2}$
   e) $y = \frac{\sqrt{1-x^2}}{x}$
   f) $y = \frac{e^x - e^- x}{2}$
   g) $y = x + \cos(x)$
9. Suppose that f and g are continuous functions on $[a, b]$ and differentiable on (a, b). Suppose also that $f(a) = g(a)$ and $f'(x) < g'(x)$ for $a < x < b$. Prove that $f(b) < g(b)$. (Hint: Apply the Mean Value Theorem (MVT) to the function $h = f - g$)
10. Does there exist a differentiable function f such that $f(0) = -1$, $f(2) = 4$, and $f'(x) \leq 2$ for all x?
11. A number a is called a **fixed point** of a function f if $f(a) = a$. Prove that if f is a differentiable function with $f'(x) \neq 1$ for all real numbers x, then f has at most one fixed point.
12. Sketch $f(x) = \sin(x) + \sqrt{3}\cos(x)$ on the interval $[-2\pi, 2\pi]$.
13. Suppose f is a continuous function where $f(x) > 0$ for all $x \in \mathbb{R}$, $f(0) = 4$, $f'(x) > 0$ if $x < 0$ or $x > 2$, $f'(x) < 0$ if $0 < x < 2$, $f''(-1) = 0 = f''(1)$, $f''(x) > 0$ if $x < -1$ or $x > 1$, $f''(x) < 0$ if $-1 < x < 1$.
    a) Can f have an absolute maximum? If so, sketch a possible graph of f. If not, explain why.
    b) Can f have an absolute minimum? If so, sketch a possible graph of f. If not, explain why.

c) Sketch a possible graph for f that does *not* achieve an absolute minimum.

14. Let n be a positive integer, then
    a) Prove that $\lim\limits_{x\to\infty} \frac{e^x}{x^n} = \infty$
    b) Prove that $\lim\limits_{x\to\infty} \frac{\ln(x)}{x^n} = 0$
    c) What can you say about the behaviour of e^x and $\ln(x)$ at large values in comparison to x^n?
15. Sketch the following curves:
    a) $y = x^{\frac{5}{3}} - 5x^{\frac{2}{3}}$
    b) $y = \frac{\sin(x)}{1+\cos(x)}$
    c) $y = \frac{\sin(x)}{2+\cos(x)}$
    d) $y = e^{-x}\sin(x)$ on interval $[0, \infty)$
16. Evaluate
$$\lim_{x\to\infty}\left(x - x^2\ln\left(\frac{1+x}{x}\right)\right)$$
17. Let
$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$
    a) Use the definition of the derivative to find $f'(0)$.
    b) Show that f has derivatives of all order, that is, $f^{(n)}(x)$ defines a function $f^{(n)}$ on $\mathbb{R}$ for each positive integer n. [*Hint:* First show by induction that for each nonnegative integer n there is a polynomial $p_n(x)$ and a nonnegative integer k_n such that $f^{(n)}(x) = \frac{p_n(x)f(x)}{x^{k_n}}$ for $x \neq 0$.]

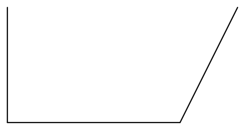
Additional problem(s)

1. Dr Shark has made himself very unpopular by setting an impossibly difficult class test. As he is leaving his office, he hears an angry mob of students approaching and he jumps on his trusty bicycle. He plans to shake off his pursuers by crossing a nearby lake in a getaway boat he has left on the one shore for such eventualities. We'll assume that the lake is circular with radius 2 km. The boat is at point A on the shore, and a friend is waiting for him in a car at point B on the opposite shore. A and B lie on a diameter of the circular lake. Dr Shark decides to go from A to B by rowing across the lake from A to another point C on the shore, and then walking from C to B. He can row at 2 km/h and walk at 4 km/h. To which point C should he row to make the total time for the trip as short as possible?
[*Hint:* C is determined by the angle CAB; call it θ. Draw a sketch of the lake, indicating A, B and C. It may be helpful to draw the line joining C to the center O of the circle.]
Remember to check that you have found the θ that gives you the minimum time, not the maximum time!

Tutorial Questions 13

Drawing a diagram for optimization problems is advisable as this makes the problem easier than it looks when written in word form.

1. You are asked to make a gutter out of a flat sheet of metal 30cm wide. The gutter has to be 10cm deep. The gutter is made by bending the sheet in the shape shown in the diagram below, which shows a cross-section of the gutter. How should you bend the sheet so that the gutter will contain as much water as possible?

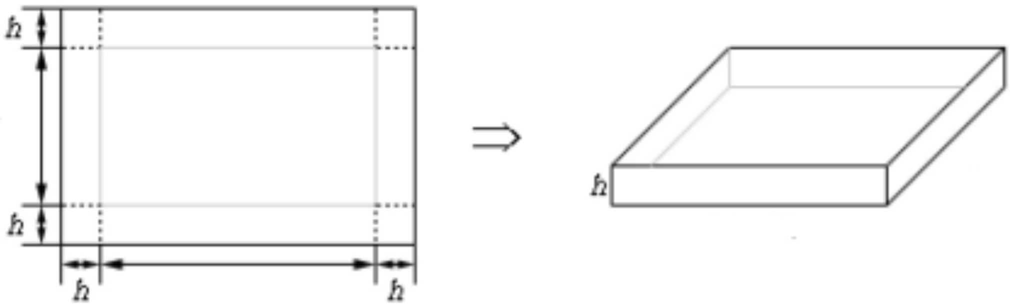


   a) At first sight, this looks as though it's a problem in 3 dimensions (since volume is a 3 dimensional idea.) In fact the problem we are looking at can be reduced to a 2 dimensional problem. Explain why. What 2 dimensional quantity must you maximize?
   b) What dimensions do you have to know before you can bend the sheet into the best shape, i.e. the shape that will carry the most water? (What are the unknowns, or variables?)
   c) Assign symbols to all the variables you mentioned in (b).
   d) The variables you listed in (c) may be related to one another. Write down any equation(s) you can think of that expresses these relationships.
   e) Write down an expression for the quantity in (a) in terms of the variables you listed in (c).
   f) The expression you wrote down may contain several variables. Try to use the equation(s) in (d) to eliminate all but one of the variables. You should now have the quantity mentioned in (a) as a function of just one variable.
   g) What values of this one variable make sense for this problem? Put differently, if we want to maximize the function, what interval of values for the variable should we consider?
   h) Differentiate the function you obtained in (f) and put the derivative equal to 0 to obtain the critical point(s) of the function in this interval.
   i) For each critical point, determine whether it gives a local maximum, minimum or neither. There are two ways of doing this; try both.
   j) Find the values of the function at the endpoints of the interval obtained in (g). Use this to find the absolute maximum and minimum of the function on the interval.
   k) You should now be able to answer the original question. A few calculations still need to be made; use your answers in (b) and (d) for this.

2. A field needs to be enclosed with a rectangular fence. If there is only 500 m of fencing material and a building is on one side of the field and so won't need any fencing. Determine the dimension of the field that will enclose the largest area.
3. A manufacturer needs to make a cylindrical can that will hold 1.5 liters of liquid. Determine the dimensions of the can that will minimize the amount of material used in its construction.
4. Find all antiderivatives of the functions given by:
   (Check your answers by differentiating.)
   a) $y = 4x + 9$
   b) $y = 2x^3 + \frac{2}{3}x^2 + 3x$

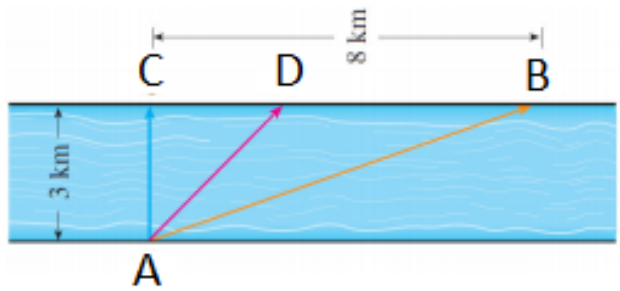
c) $y = 5\sqrt{x} - 2\sqrt[3]{x}$
d) $y = x(2-x)^2$
e) $y = 2\sin(x) - \sec^2(x)$
f) $y = 2\sqrt{x} - \sec(x)\tan(x)$

5. Given a piece of cardboard that is 140 cm by 100 cm. Determine the height of the box that will give a maximum volume if the box is constructed by cutting out the corners and folding up the sides as shown below.



6. A printer needs to make a poster that will have a total area of 400 cm^2 and 2 cm margins on the sides, a 4 cm margin on top and a 1.5 cm margin on the bottom. What dimensions of the poster will give the largest printed area?
7. For each of the following, find f.
   a) $f'(x) = \frac{x+1}{\sqrt{x}}$, and $f(1) = 5$
   b) $f'(x) = x - \frac{1}{x^3}$, $x > 0$ and $f(1) = 6$
   c) $f'(x) = \sec(x)[\sec(x) + \tan(x)]$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$ and $f(\frac{\pi}{4}) = -1$
8. For each of the following, find f.
   a) $f'(x) = \frac{2}{1+x^2}$, and $f(1) = \frac{\pi}{4}$.
   b) $f''(x) = -2 + 12x - 12x^2$, and $f(0) = 3$, $f'(0) = 12$.
   c) $f''(x) = \sin(x) + \cos(x)$, and $f(0) = 3$, $f'(0) = 4$
   d) $f''(x) = \sqrt[3]{x} - \cos(x)$, and $f(0) = 2$, $f'(1) = 2$
9. A window is being built and the bottom is a rectangle and the top is a semicircle. If there is 16 m of framing material, what must be the dimensions of the window to let in the most light?
10. Find a function f such that $f'(x) = x^3$ and the line given by $x + y = 0$ is a tangent to f at some point. (Is this function unique?)
11. Determine the points on $y = x^2 + 1$ that are closest to $(0, 2)$.

12. A man launches his boat from point A on a bank of a straight river which is 3 km wide and wants to reach to point B, 8 km downstream on the opposite bank, as quickly as possible. If he can row at 6 km/h and run at 8 km/h, Find the point D where he should land and run to point B so to minimize time spent on a trip from A to B.



13. ABCD is a square piece of paper with sides of length 1 m. A quarter-circle is drawn from B to D with center A. The piece of paper is folded along EF, with E on AB and F on AD, so that A falls on the quarter-circle. Determine the maximum and minimum areas that the triangle AEF can have. [Hint: Draw the diagram.]
14. An oil tanker has developed a leak in one of its tanks. During the first hour oil leaks from the tank at a rate of $r(t) = \dfrac{10}{1+t^2}$ tonnes per hour, where t is measured in hours from the time the leak started.
    a) Write down expressions giving the amount of oil that leaked from the tanker (i) in the first 15 minutes (ii) between 30 and 60 minutes after the leak started. You need not evaluate these expressions.
    b) If the expression in (a)(i) is approximated by a right-hand sum, will it be an under- or over-estimate? Give a reason for your answer
    c) Give an anti-derivative for the function $r(t) = \dfrac{10}{1+t^2}$.
    d) Use your answer to (c) to evaluate the expressions in (a).
    e) Thirty minutes after the leak started, an anti-pollution team starts spraying a chemical solvent on the oil slick in an attempt to disperse it. The solvent removes oil from the sea at a constant rate of 2 tonnes per hour. Write down an expression for the amount of oil in the sea after 1 hour and evaluate it.
    f) If oil continues leaking from the tanker at a rate given by $r(t)$ after the first hour, and the dispersant is sprayed at the same constant rate, when will the amount of oil in the sea reach a maximum? What is this maximum amount?
15. a) Use Newton's method to find the fifth root of 3. ie. Find x for $x = 3^{\frac{1}{5}}$. Hint: rearrange this to give you a function whose zero you are trying to find.
    b) One can tell from inspection that $x = 2$ and $x = 4$ are solutions to equation $2^x = x^2$. Use Newton's method to find the third solution.

Tutorial Questions 14

1. Write the following sums in sigma notation:
   a) $S = 2 + 3 + 4$
   b) $S = 1^2 + 2^2 + 3^2 + 4^2$
   c) $S = 4 + 7 + 10 + 13$
   d) $S = (4+5)^2 + (5+6)^2 + (6+7)^2 + (7+8)^2$
   e) $S = 1 - 2 + 3 - 4 + 5 - 6 + 7 - 8$
   f) $S = (1 + 2 + 3 + 4) - (1 + 4 + 9 + 16)$

2. First expand, then evaluate the following sums given in sigma notation (note that sometimes we can collect together terms to make it easier. (For instance $1+2+3+4+5+6 = (1+6)+(2+5)+(3+4) = 7+7+7 = 3\times 7$). For the sum with exponentials, you don't need to evaluate the total.:

   (a) $\displaystyle\sum_{n=1}^{6} n^2$ (b) $\displaystyle\sum_{k=1}^{4} \frac{1}{k^2}$ (c) $\displaystyle\sum_{n=1}^{5} n$ (d) $\displaystyle\sum_{n=1}^{17} 1$

   (e) $\displaystyle\sum_{n=1}^{3} e^{2n}$ (f) $\displaystyle\sum_{n=1}^{1} 17$ (g) $\displaystyle\sum_{n=1}^{4} \left[\frac{n(n+1)}{2}\right]^2$ (h) $\displaystyle\sum_{k=1}^{4} \frac{k(k+1)(2k+1)}{6}$

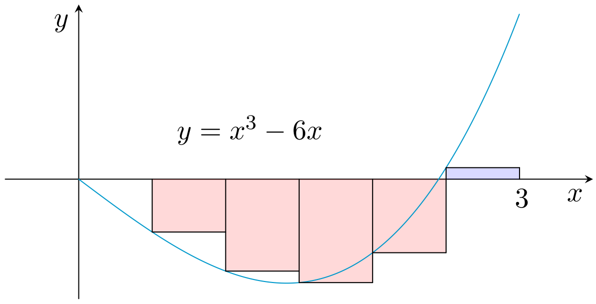
3. Prove the following results:

   (a) $\displaystyle\sum_{k=m}^{n} ca_k = c\sum_{k=m}^{n} a_k$ (b) $\displaystyle\sum_{k=m}^{n} (a_k + b_k) = \sum_{k=m}^{n} a_k + \sum_{k=m}^{n} b_k$ (c) $\displaystyle\sum_{k=m}^{n} (a_k - b_k) = \sum_{k=m}^{n} a_k - \sum_{k=m}^{n} b_k$

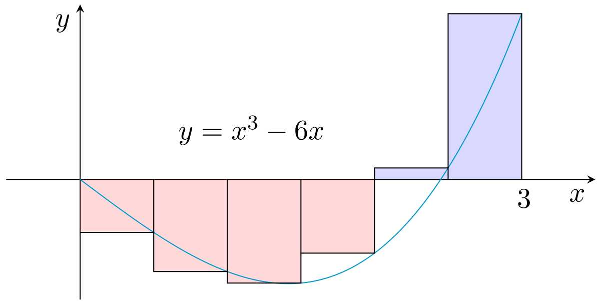
   (d) $\displaystyle\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ (e) $\displaystyle\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ (f) $\displaystyle\sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2$

   Two of the last three of these are explained in detail here: http://www.mathemafrica.org/?p=13557

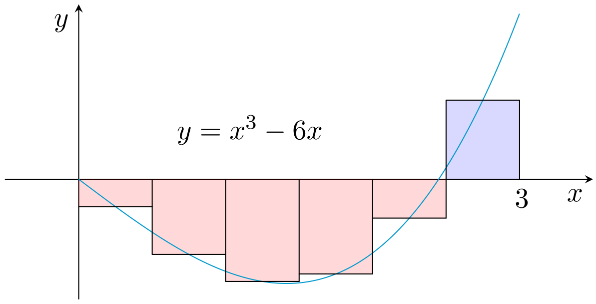
4. For the diagrams below with Riemann sum approximations shown (all are drawn to scale):
(i) Identify $x_0, x_1, x_2, x_3, x_4, x_5$ and x_6[1] Hence, identify Δx.
(ii) Each diagram is labeled with the rule to be used for the Riemann sum. Identify all the sample points x_i^* and find $f(x_i^*)$ for all x_i^* in part (i).
(iv) Write down and evaluate an expression that approximates the area between the curve and the x-axis using the rectangles as the approximation.
(v) Write down an expression that approximates the area between the curve and the x-axis and holds for any general function and the type of sample points (left endpoints, right endpoints, etc.) you are using. [*Hint:* You should use sigma notation.]



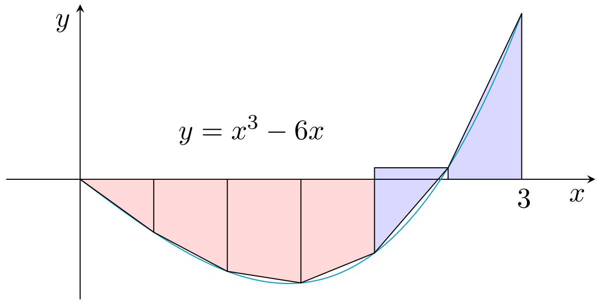
(a) Left endpoints



(b) Right endpoints



(c) Midpoints



(d) Trapezoidal

5. Use the left, right and midpoint rules with the given number of subintervals, n, to approximate the following integrals. What do you notice?

a) $\displaystyle\int_2^{10} x^2 dx, \quad n = 4$

b) $\displaystyle\int_0^{5} (1 + 2x^3) dx, \quad n = 5$

6. Intuitively, for a well-behaved function, we expect our approximations to get better as we let $n \to \infty$ regardless of which sampling points we use. Write the following Riemann sums as definite integrals.

(a) $\displaystyle\lim_{n\to\infty} \sum_{i=1}^{n} x_i \ln(1 + x_i)\Delta x, \quad [2, 6]$

(b) $\displaystyle\lim_{n\to\infty} \sum_{i=1}^{n} \frac{\cos x_i}{x_i}\Delta x, \quad [\pi, 2\pi]$

(c) $\displaystyle\lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{2x_i^* + (x_i^*)^2}\Delta x, \quad [1, 8]$

(d) $\displaystyle\lim_{n\to\infty} \sum_{i=1}^{n} \left[4 - 3(x_i^*)^2 + 6(x_i^*)^5\right] \Delta x, \quad [0, 2]$

[1] In the previous version, this was x_1 to x_7 which is consistent but different from the notation you may have used before.

7. Explain why the following is true:

$$\lim_{n\to\infty}\sum_{i=1}^{n} x_i \ln(1+x_i)\Delta x, \quad [2,6] \equiv \lim_{n\to\infty}\sum_{i=1}^{n}\left(2+\frac{4(i-1)}{n}\right)\ln\left(1+2+\frac{4(i-1)}{n}\right)\frac{4}{n}$$

and show how you would write the other Riemann sums in question 6 in the same way. In the previous version, there were factors of $n-1$. This was correct, but less clear. Why are the limits the same?

8. There are eight properties of definite integrals given in Stewart on pages 313-315, write them all down and prove the first four. You will need to prove them by writing each one as a Riemann sum and using the properties of sums to prove the integral properties.

9. Express the following integrals as Riemann sums.

(a) $\displaystyle\int_2^6 \frac{x}{1+x^5}dx$ (b) $\displaystyle\int_1^{10}(x-4\ln x)dx$

(c) $\displaystyle\int_0^{\pi}\sin 5x dx$ (d) $\displaystyle\int_2^{10} x^6 dx$

10. **Combining the Chain Rule with FTC1**. Evaluate the following integrals.

(a) $\displaystyle\frac{d}{dx}\int_1^{x^4}\sec t dt$ (b) $\displaystyle\frac{d}{dx}\int_{\tan x}^{3}\sin(y^2)dy$ (c) $\displaystyle\frac{d}{dx}\int_{-\sqrt{x^3}}^{1/x}\frac{1}{1+y^2}dy$

11. Solve the following definite integrals.

(a) $\displaystyle\int_2^4 \sec^2 t dt$ (b) $\displaystyle\int_0^{\frac{\pi}{4}}\frac{1}{-\sqrt{1-x^2}}dx$ (c) $\displaystyle\int_{-\pi/2}^{\pi/2}\frac{1}{1+y^2}dy$ (d) $\displaystyle\int_{-1}^{4} a^x dx$

12. Prove (without using a calculator!) that

$$\frac{1}{4}\leq\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\frac{\sin x}{x}\,dx\leq\frac{\sqrt{3}}{2}.$$

[*Hint:* Do not try to evaluate the integral.]

13. The function

$$f(x)=\frac{1}{\sqrt{2\pi}}e^{-x^2/2}$$

describes the *standard normal distribution*, probably the most frequently used distribution in statistics. Definite integrals of this function are used in statistical tests. The following table gives the values of a few such integrals:

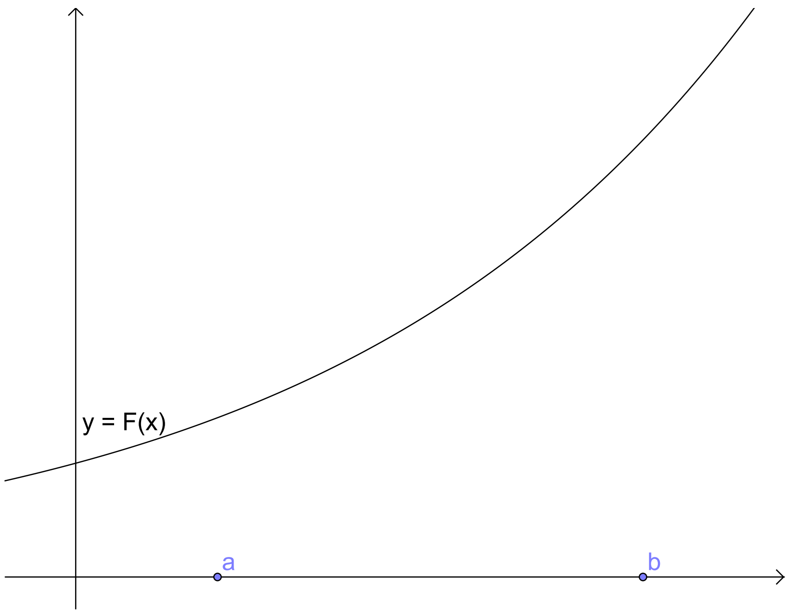
b	$\int_0^b f(x)\,dx$
1	0.3413
2	0.4772
3	0.4987
4	0.5000

Use the table to find the value of each of the following integrals:

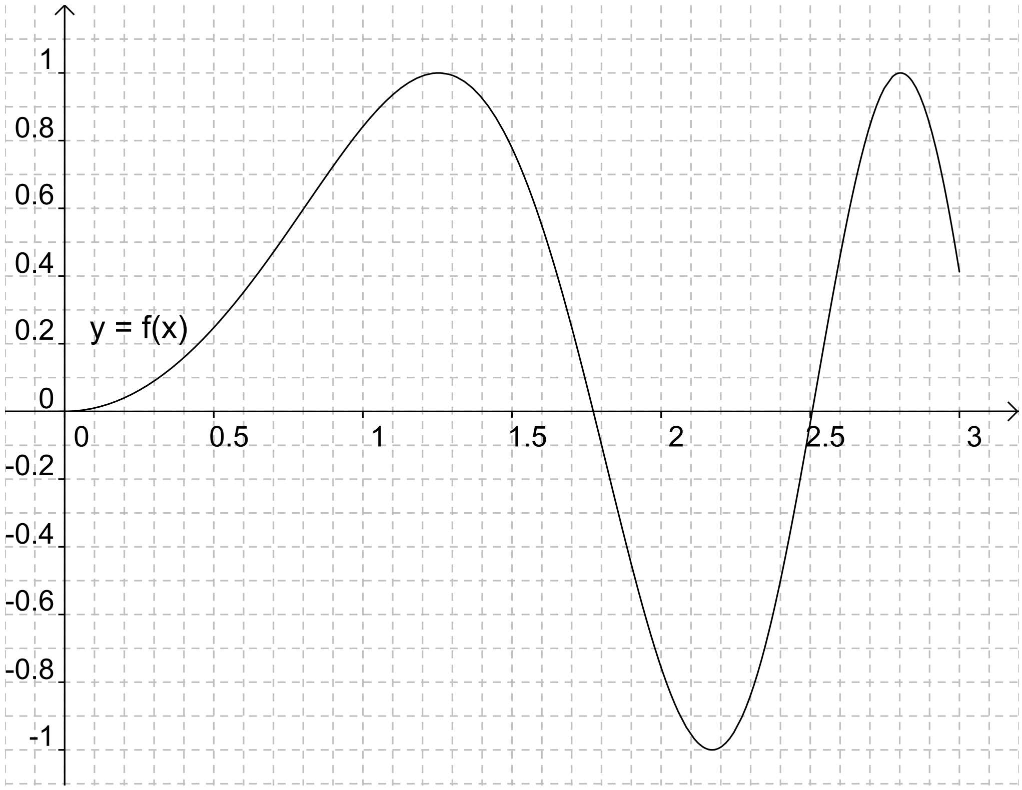
(a) $\displaystyle\int_1^3 f(x)\,dx$ (b) $\displaystyle\int_{-1}^{0} f(x)\,dx$

(c) $\displaystyle\int_0^{-4} f(x)\,dx$ (d) $\displaystyle\int_{-2}^{3} f(x)\,dx$

14. Let F be an antiderivative of f. On the graph of F shown below, indicate
   a) a slope representing $f(b)$;
   b) a length representing $\displaystyle\int_a^b f(x)\,dx$;
   c) a slope representing $\dfrac{1}{b-a}\int_a^b f(x)\,dx$;
   d) a slope representing $\displaystyle\lim_{h\to 0}\frac{1}{h}\int_a^{a+h} f(x)\,dx$.



15. The graph of the function f on the interval $[0,3]$ is shown below. Let F be the antiderivative of f for which $F(0)=1$.
In this question you have to find various approximations. Rough approximations will do; **do not** *spend a lot of time trying to get accurate answers.*
   a) Use a left-hand sum and a right-hand sum, in both cases with a division into 10 subintervals, to find approximations to $\displaystyle\int_0^1 f(x)\,dx$.
   b) Find an approximate value for $F(1)$.
   c) List the numbers $F(1.5), F(2), F(2.5)$ in increasing order. (You need not find approximations to them.)
   d) Does the function F have any local maximum or minimum values on the interval $[0,3]$? If so, find the approximate x-coordinate of each such point and say whether it is a maximum or minimum.
   e) For which value of x does F attain its absolute maximum value on $[0,3]$? Find this maximum value approximately.
   f) What is the absolute minimum value of F on $[0,3]$ and where is it attained?
   g) Does the graph of F have any points of inflection on $[0,3]$? If so, give the (approximate) x-coordinates of each of these points.


y = f(x)
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
0.5
1
1.5
2
2.5
3

Tutorial Questions 15

1. For each of the following statements, say whether it is true or false. If it is true, explain why; if it is false, give the correct statement.
   [*Hint:* One can check whether a given indefinite integral is correct without having to integrate.]
   a) $\displaystyle\int \frac{1}{1+e^u}\,du = u - \ln(1+e^u) + C$
   b) $\displaystyle\int_1^3 \frac{x^2}{\sqrt{x^3+8}}\,dx = \frac{1}{3}\int_9^3 5\frac{du}{\sqrt{u}}.$
   c) $\displaystyle\int \tan^n t\,dt = \frac{\tan^{n-1} t}{n-1} - \int \tan^{n-2} t\,dt$
   [*Hint:* You may have to use a trig identity here.]
2. Consider the integral $\displaystyle\int \cos 2x dx$.
   a) Which differentiation rule is integration by substitution the inverse of?
   b) Using the substitution $u = 2x$, find an expression to replace dx and interpret what this says in words.
   c) Using the expressions for $2x$ and dx, rewrite and solve $\displaystyle\int \cos 2x dx$.
   d) Evaluate $\displaystyle\int_{-\pi}^{\pi} \cos 2x dx$ and $\displaystyle\int_{-\pi}^{\pi} \sin 2x dx$.
   e) What can be said about $\displaystyle\int_{-a}^{a} f(x)dx$ when f is even and when f is odd?
   f) Using your answer to part (e), re-evaluate your answers in part (d).
3. Solve the following integrals using substitution.
   (a) $\displaystyle\int x^2\sqrt{x^3+1}dx$ (b) $\displaystyle\int \sin^2\theta\cos\theta d\theta$ (c) $\displaystyle\int \frac{x^3}{(x^4-5)^2}dx$
   (d) $\displaystyle\int \sqrt{2t+1}dt$ (e) $\displaystyle\int \tan x dx$
4. Consider the integral $\displaystyle\int x\sin x dx$.
   a) Which differentiation rule is integration by parts the inverse of?
   b) Show that $\displaystyle\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx$
   c) For the considered integral, what are sensible choices for $f(x)$ and $g'(x)$?
   d) Rewrite $\displaystyle\int x\sin x dx$ using part (b) and solve for it.
   e) Evaluate $\displaystyle\int_{-\pi}^{\pi} x\sin x dx$ [*Hint:* Consider whether $f(x) = x\sin x$ is even or odd.]
   f) Hence or otherwise evaluate $\displaystyle\int_{-\pi}^{2\pi} x\sin x dx$.
5. Solve the following integrals by parts.
   (a) $\displaystyle\int \ln x dx$ (b) $\displaystyle\int t^2e^t dt$ (c) $\displaystyle\int e^{2x}\sin x dx$
   (d) $\displaystyle\int \arctan x dx$ (f) $\displaystyle\int x^m\cos x,\quad m\in\mathbb{Z}^+$

6. **Extra problems from Stewart.** These are a mixture of integration by substitution and integration by parts problems. (See Stewart 5.5 and 5.6).

6.1. Suppose f is continuous on $[-a, a]$. Show that

(a) If f is even, then $\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx$

(b) If f is odd, then $\int_{-a}^{a} f(x)dx = 0$

(c) Solve the following integrals:

(i) $\int_{-\pi/2}^{\pi/2} \frac{x^2 \sin x}{1+x^6}dx$ (ii) $\int_{-\pi/4}^{\pi/4} \left(x^3 + x^4 \tan x\right) dx$ (iii) $\int_{0}^{a} x\sqrt{a^2-x^2}dx$

6.2. (a) Prove the reduction formula

$\int \sin^n xdx = -\frac{1}{n}\cos x sin^{n-1}x + \frac{n-1}{n}\int sin^{n-2}xdx$ where $n \geq 2$ is an integer.

(b) Evaluate $\int \sin^6 5xdx$.

6.3. (a) Use integration by parts to show that

$\int f(x)dx = xf(x) - \int xf'(x)dx$.

(b) If f and g are inverse functions and f' is continuous, prove that

$\int_{a}^{b} f(x)dx = bf(b) - af(a) - \int_{f(a)}^{f(b)} g(y)dy$.

(c) Use part (b) to evaluate $\int_{1}^{e} \ln xdx$.

6.4. A particle that moves along a straight line has velocity $v(t) = t^2e^{-t}$ meters per second after t seconds. How far will it travel during the first t seconds?

6.5. Suppose that $f(1) = 2, f(4) = 7, f'(1) = 5, f'(4) = 3$ and f'' is continuous. Find the value of $\int_{1}^{4} xf''(x)dx$.

6.6. Solve the following additional problems

(a) $\int \frac{(\ln x)^2}{x}dx$ (b) $\int \frac{dx}{5-3x}$ (c) $\int \frac{x}{(x^2+1)^2}dx$

(d) $\int x(2x+5)^8dx$ (e) $\int \cos\sqrt{x}dx$ (f) $\int e^{\cos t}\sin tdt$

(g) $\int \frac{x\cos 2x}{2\cot 2x}dx$ (h) $\int \frac{x^2}{x^2+4}dx$ (i) $\int \frac{\arctan x}{x^2+1}dx$

(j) $\int \frac{2x\sin x}{\cos^3 x}dx$ (k) $\int_{0}^{T/2} \sin\left(2\pi t/(T-\alpha)\right)dt$ (l) $\int \frac{1}{\sqrt{1-t^2}}dt$

(m) $\int_{0}^{1} \frac{dx}{(1+\sqrt{x})^4}$ (n) $\int_{e}^{e^4} \frac{dx}{x\sqrt{lnx}}$

7. Find the following integrals. In each case there is a quick way of doing it; try to find it.

a) $\int_{-1}^{1} \tan^3 x\, dx$

b) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |\sin x|\, dx$

c) $\int_0^1 x\sqrt{1-x^4}\, dx$ [*Hint:* Try to obtain an integral that you can interpret as the area of something familiar.]

d) $\int_{-1}^1 x\sqrt{1-x^4}\, dx$

8. a) Prove that for every continuous function f

$$\int_0^{\frac{\pi}{2}} f(\sin x)\, dx = \int_0^{\frac{\pi}{2}} f(\cos x)\, dx.$$

b) Use (a) to find $\int_0^{\frac{\pi}{2}} \sin^2 x\, dx$ and $\int_0^{\frac{\pi}{2}} \cos^2 x\, dx$.

9. Prove the following reduction formulas:

a) $\int (\ln x)^n = x(\ln x)^n - n\int (\ln x)^{n-1}\, dx.$

b) $\int x^n e^x = x^n e^x - n\int x^{n-1} e^x dx.$

10. Let $I_n = \int_0^{\pi/2} sin^n x dx.$

a) Show that $\int_0^{\pi/2} sin^{2n} x dx = \frac{1\cdot 3\cdot 5\cdot \cdots \cdot (2n-1)}{2\cdot 4\cdot 6\cdot \cdots \cdot 2n}\frac{\pi}{2}.$

b) Show that $I_{2n+2} \leq I_{2n+1} \leq I_{2n}$.

c) Using part (a), show that $\frac{I_{2n+2}}{I_{2n}} = \frac{2n+1}{2n+2}$.

d) Use parts (b) and (c) to show that $\frac{2n+1}{2n+2} \leq \frac{I_{2n+1}}{I_{2n}} \leq 1$ and deduce that

$$\lim_{n\to\infty} \frac{I_{2n+1}}{I_{2n}} = 1$$

.

11. Solve for the following definite integrals (You have to decide what techniques will be most efficient):

(a) $\int_1^e \sin(\ln x)\, dx$ (b) $\int_0^{33} x^2 \sqrt[5]{x-1} dx$ (c) $\int_0^{\pi/4} \frac{\sec^2 x}{1+\tan x} dx$

(d) $\int_1^2 \frac{\ln x}{x^2} dx$ (e) $\int_{-\pi/4}^{\pi/4} \sec^2 x dx$ (f) $\int_{-\pi/4}^{\pi/4} \tan^2 x dx$

12. Solve $\int \frac{1}{t+\sqrt[3]{t}+2\sqrt[3]{t^2}} dt \qquad t > 0.$

Hint: Use the substitution $u = t^{1/3}$. After this you can either make another substitution or use integration by parts. You will find that you get different answers these ways...or at least it seems so. Why are they actually the same answer?